
Designed Experiments

A carefully planned investigation can give much more insight into the question at hand than a haphazard one, data from which may be useless. Experimental design is a highly developed subject, though its principles are not universally appreciated. In this chapter we outline some basic ideas and describe some simple designs and associated analyses. The first section discusses the importance of randomization, and shows how it can be used to justify standard linear models and how it strengthens inferences. Section 9.2 then describes some common designs and analyses. Interaction, contrasts and analysis of covariance are discussed in Section 9.3. Section 9.4 then outlines the consequences of having more than one level of variability.

9.1 Randomization

9.1.1 Randomization

The purpose of a designed experiment is to compare how *treatments* affect a response, by applying them to experimental *units*, on each of which the response is to be measured. The units are the raw material of the investigation; formally a unit is the smallest subdivision of this such that any two different units might receive different treatments. The treatments are clearly defined procedures one of which is to be applied to each experimental unit. In an agricultural field trial the treatments might be different amounts of nitrogen and potash, while a unit is a plot of land. In a medical setting, treatments might be types of operation and different therapies, with units being patients who are operated upon and then given therapy to aid recovery. In each case our concern is how the response depends on the treatment combinations and other measurable quantities. The response must be carefully defined and measured in a consistent way for every unit.

Suppose for illustration that we wish to assess the effect of a drug in reducing blood pressure, and that $n = 2m$ individuals are available. We plan to administer the drug to m of the individuals, the treatment group, and to give a placebo to the remaining m ,

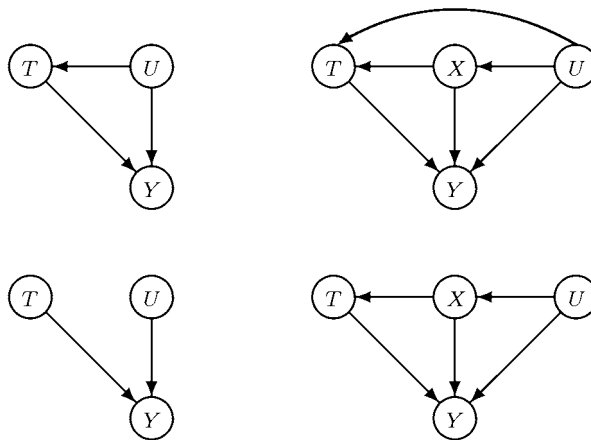


Figure 9.1 Directed acyclic graphs showing consequences of randomization. An arrow from T to Y indicates dependence of Y on T , and so forth. In general both response Y and treatment T may depend on properties U of units (upper left). Randomization (lower left) makes treatments and units independent, so any observed dependence of Y on T cannot be ascribed to joint dependence on U . The upper right graph shows the general dependence of Y , T , and covariates X on U . Randomization makes T and U independent, conditional on X (lower right), so any influence of U on T is mediated through X , for which adjustment is possible in principle. Thus having adjusted for X , dependence of Y on T cannot be due to U .

the control group. The response is to be the blood pressure of an individual measured a fixed time after the drug has first been administered. We calculate the average changes for the treated and control groups, $\bar{y}_1$ and $\bar{y}_0$, observe that $\bar{y}_1 - \bar{y}_0$ is significantly less than zero, and declare that the drug plays an effect in reducing blood pressure. Is this headline news? No!

A key difficulty is that the procedure does not avoid biased allocation of treatments to units. For example, if the control group mostly consisted of those patients with higher blood pressures at the start of the study, $\bar{y}_1$ and $\bar{y}_0$ might differ greatly even if the treatment had been ineffective. This particular source of bias could be avoided if the experimenter measured the initial blood pressures and deliberately balanced the groups with respect to them, but unknown causes of bias could not be removed in this way, and the interpretation of the results would rely on the uncheckable assertion that the experiment was also balanced with respect to these unknown factors. Any deterministic allocation scheme will have this flaw, and we turn instead to *randomization*. By allocating treatments to patients at random, we expect to equalize the effect of any factors that might affect the response, other than the treatment itself. We can then be surer that a significant difference between the groups is related to the treatment itself.

To explain randomization differently, let T represent the treatment, Y the response, and U properties of units — potential sources of bias. For example, left to their own devices physicians might be tempted to allocate a promising but untested new treatment to patients most severely affected by a disease, and an existing treatment to less severe cases. Then treatment T would depend on an attribute of the units, disease severity U ; the response Y might depend on both T and U . This is shown by the directed acyclic graph in the upper left part of Figure 9.1. In general both T and Y depend on U , so any apparent relation between Y and T may be ascribed to U . Randomization induces independence between properties of the unit and any treatment allocation, making T independent of U and the lower left graph appropriate: although U may influence the response Y , it cannot entirely explain any dependence on T unless the randomization is compromised, for example by allocating all men to one group and all women to the other purely by chance. If this has not happened,

Allocation at random means that some physical device has been used, not that the experimenter has made a choice that appears haphazard.

then a highly significant effect of T implies either that treatment works, or that a rare event has occurred.

If randomization had been used and if a normal linear model was suitable, inference could be based on the two-sample model of Example 8.9, using

$$z = \frac{\bar{y}_1 - \bar{y}_0}{(2s^2/m)^{1/2}}, \quad (9.1)$$

where $s^2 = (2m - 2)^{-1} \sum_{t,j} (y_{tj} - \bar{y}_t)^2$ is the pooled estimate of error and y_{tj} is the response for the j th individual in treatment group t . In fact randomization gives a basis for the use of this and other linear models, as we shall see below.

Blocking

The design outlined above presupposes that the units are fairly homogeneous, that is, any variation among blood pressures of different patients is small enough for the design to be *completely randomized*. However, if the treatment effect was small relative to this variation, s^2 would be inflated because the division into groups made no allowance for it. The larger is s^2 , the smaller is z for given $\bar{y}_1 - \bar{y}_0$, and this makes it harder to detect any treatment effect. This suggests that we should subdivide the patients into groups whose initial blood pressures are as alike as possible, and allocate the treatment randomly within these groups, a procedure known as *blocking*. As the purpose of our experiment is to compare one treatment with the control, we divide the patients into m blocks of two individuals with similar initial blood pressures, and randomly allocate one of each pair to the treatment and the other to the control, in a *paired comparison*. In the corresponding normal linear model, discussed in Example 8.10, analysis is based on the differences d_j between the treated and control individuals in the j th block, leading to confidence statements using the standardized difference given by

$$z_d = \frac{\bar{d}}{(s_d^2/m)^{1/2}}, \quad s_d^2 = (m - 1)^{-1} \sum_{j=1}^m (d_j - \bar{d})^2, \quad (9.2)$$

where $\bar{d} = \bar{y}_1 - \bar{y}_0$ is the average difference between pairs. The numerator of z_d is the same as that of (9.1), but the denominator may be substantially smaller if the blocking has been effective in increasing the precision of the experiment. Although here the matching is performed deliberately, randomization is still involved in the treatment allocations.

This line of reasoning suggests taking as response for each patient the difference between his initial blood pressure and that after treatment, so the comparisons are made entirely within individuals, allocated randomly to treatment or control. We ignore this design below, however, purely for purposes of exposition.

The right half of Figure 9.1 shows the effect of randomization when treatment allocation can depend on a covariate, X . For example, randomization might take into account knowledge that certain treatments should not be given to patients taking other medication. In general T might depend on unknown properties U of the unit as well as on X , so that Y and T depend on both X and U . Randomization breaks the direct

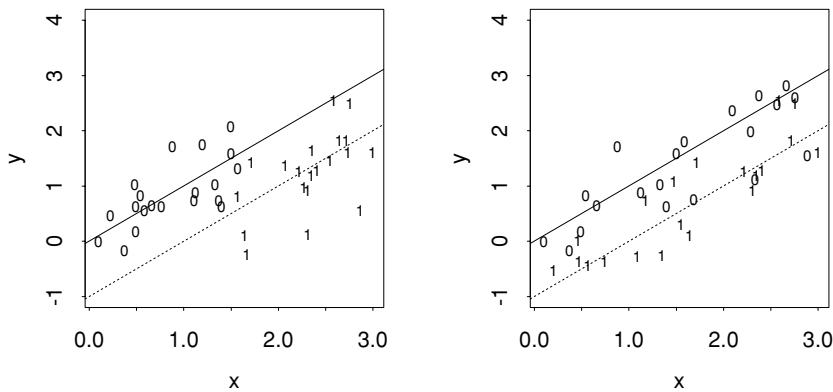


Figure 9.2 Simulated results from experiments to compare the effect of a treatment T on a response Y that varies with a covariate X . The lines show the mean response for $T = 0$ (solid) and $T = 1$ (dots). Left: the effect of T is confounded with dependence on X . Right: the experiment is balanced, with random allocation of T dependent on X .

link between U and T , so any effect of X on T is mediated through the observed X , for which adjustment is in principle possible.

To illustrate this, Figure 9.2 shows results from two simulated attempts to assess the effect of a treatment T on a response Y . Unnoticed by the virtual experimenter who obtained the data in the left panel, the mean of Y increases with a covariate X , as shown by the lines. However because all the units for which $T = 1$ also have the largest values of X , there appears to be no difference between the treatment group averages. The true treatment effect is $\delta = -1$, but the observed difference of averages is 0.2 with standard error 0.2. The 0.95 confidence interval $(-0.2, 0.6)$ does not include the true δ because of *confounding* between the effects of X and T . In practice such serious confounding would be most likely to arise due to lack of randomization, but lack of balance could occur by accident even if the treatments had been allocated at random. If so, randomization would fail to remove all possible biases due to confounders such as X .

A canner experimenter might have formed pairs of units using values of X measured before the experiment and then randomized the treatment within pairs, leading to results like those in the right panel, where the difference of averages is -1.2 with standard error 0.3; the 0.95 confidence interval now contains δ .

In both cases the observed values of X can be used to obtain more precise estimates of δ , by fitting the model $y = \beta_0 + \beta_1 x + \delta t + \varepsilon$ to the observed triples (x, t, y) , where $t = 0$ or 1. The left panel has $\hat{\delta} = -0.7$ with standard error 0.3 and correlation $\text{corr}(\hat{\delta}, \hat{\beta}_1) = -0.82$, while the right has $\hat{\delta} = -1.25$, standard error 0.16 and $\text{corr}(\hat{\delta}, \hat{\beta}_1) = -0.04$. One effect of the blocking has been to reduce the confounding of T and X by making the corresponding columns of the design matrix almost orthogonal; their parameters can then be estimated without ambiguity. There is a relation here to the discussion of collinearity in Section 8.7.2.

Although regression on x reduces the confounding between X and T in the first experiment, the lack of overlap in the values of X for the two treatment groups means that the model must be used to interpolate between them. This makes the estimate less precise and the inference less secure: an act of faith in the linearity of the model is needed, because neither of the groups has X values over the entire range.

The second experiment gives similar estimates of δ with or without adjustment for x , though the precision of $\hat{\delta}$ is increased by making the adjustment, known as analysis of covariance; see Section 9.3.3. Moreover the data can be used to check whether the treatment effect is constant over X .

Randomization inference

In this chapter we shall assume that normal linear models are applicable. In fact the act of randomization provides a basis for inference without appealing to specific parametric assumptions, but for which the normal model often provides a good approximation. Suppose that m observations have been randomly allocated to a treatment and a further m to a control. Suppose also that *unit-treatment additivity* holds, that is there exist constants $\gamma_1, \dots, \gamma_{2m}$, one for each unit, and δ for the treatment, such that the response on the j th unit is $\gamma_j + \delta$ when it is allocated to the treatment, and $\gamma_j - \delta$ if it is allocated to the control group, regardless of the allocation of treatments to the other units. Thus the effect of treatment is to increase the response by $\Delta = 2\delta$ relative to the control, for each unit in the experiment. Under this model the responses from the j th unit when it is allocated to treatment and to control are

$$T_j(\gamma_j + \delta), \quad (1 - T_j)(\gamma_j - \delta),$$

where T_j is an indicator of whether it has been allocated to the treatment. Therefore the difference between treatment and control averages is

$$\bar{Y}_1 - \bar{Y}_0 = \frac{1}{m} \sum_{j=1}^{2m} T_j(\gamma_j + \delta) - \frac{1}{m} \sum_{j=1}^{2m} (1 - T_j)(\gamma_j - \delta) = 2\delta + \frac{1}{m} \sum_{j=1}^{2m} (2T_j - 1)\gamma_j.$$

The properties of $\bar{Y}_1 - \bar{Y}_0$ stem from the moments of $T_1, \dots, T_{2m}$,

$$E(T_j) = \frac{1}{2}, \quad E(T_j T_k) = \frac{m-1}{2(2m-1)}, \quad j \neq k. \quad (9.3)$$

Thus $\bar{Y}_1 - \bar{Y}_0$ has mean Δ and variance $2\{m(2m-1)\}^{-1} \sum_{j=1}^{2m} (\gamma_j - \bar{\gamma})^2$. Moreover the strong symmetry induced by the T_j , allied to the weak dependence among them, means that the randomization distribution of $\bar{Y}_1 - \bar{Y}_0$ is close to normal.

Example 9.1 (Shoe data) Table 9.1 shows the amount of wear in a paired comparison of materials A and B used to sole shoes. Material B is cheaper and the aim of the experiment was to see if it was less durable than A. Ten boys were chosen, material A allocated at random to one of their shoes, and material B to the other. All but two of the differences d_j are positive, suggesting that shoes soled with B wear more quickly than those with A. The average difference is $\bar{d} = 0.41$.

Suppose that there was no difference between the materials. Then A and B would simply be labels attached randomly to the shoes, and each difference might equally well have had the opposite sign. That is, each of the $2^{10} = 1024$ outcomes $\pm 0.8, \pm 0.6, \dots, \pm 0.3$ would have been equally as likely as that actually observed. Thus the average difference $\bar{d}$ would be the observed value of $\bar{D} = m^{-1} \sum_j D_j$, where $D_j = I_j d_j$, and $I_1, \dots, I_m$ are independent variables taking values ± 1 with

Boy	Material		Difference <i>d</i>
	A	B	
1	13.2 (L)	14.0 (R)	0.8
2	8.2 (L)	8.8 (R)	0.6
3	10.9 (R)	11.2 (L)	0.3
4	14.3 (L)	14.2 (R)	−0.1
5	10.7 (R)	11.8 (L)	1.1
6	6.6 (L)	6.4 (R)	−0.2
7	9.5 (L)	9.8 (R)	0.3
8	10.8 (L)	11.3 (R)	0.5
9	8.8 (R)	9.3 (L)	0.5
10	13.3 (L)	13.6 (R)	0.3

Table 9.1 Shoe wear data (Box *et al.*, 1978, p. 100). The table shows the amount of shoe wear in an paired comparison experiment in which two materials A and B were randomly assigned to the soles of the left (L) or right (R) shoe of each of ten boys.

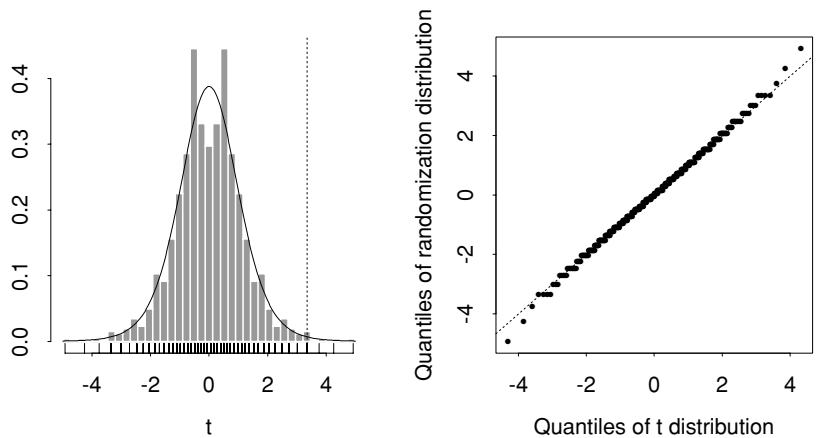


Figure 9.3 Randomization distribution of the *t* statistic for the shoes data, together with its approximating *t*₉ distribution. The left panel shows a histogram and rug for the randomized values of *Z*, with the *t*₉ density overlaid; the observed value is given by the vertical dotted line. The right panel shows a probability plot of the randomization distribution against *t*₉ quantiles.

probability $\frac{1}{2}$; here $m = 10$. In fact there are precisely three values of $\overline{D}$ that are larger than $\overline{d}$, and four values equal to it, so the exact P-value based on $\overline{D}$ is $7/1024 \doteq 0.007$.

The studentized version of $\overline{D}$, $Z = \overline{D}/[\{m(m - 1)\}^{-1} \sum (D_j - \overline{D})^2]^{1/2}$, is a monotonic function of $\overline{D}$, so both Z and $\overline{D}$ give the same P-values under randomization. Figure 9.3 shows the randomization distribution of Z , with the *t* distribution on $m - 1 = 9$ degrees of freedom that would be used under a normal model. The agreement between the randomization distribution and the normal approximation is excellent. The observed value of Z is 3.35, with significance level 0.004 when compared to the *t*₉ distribution.

The pairing in this experiment could have been used to extend the validity of the results, by taking boys of different ages, with different types of shoes and so forth. As the comparisons are based only on differences between feet of the same boy, that is, within blocks, the heterogeneity of the boys themselves does not affect the comparison of A and B. If the same difference between materials was seen on a wide variety of blocks, one could be more confident that the difference in durability was general. As previously mentioned, blocking is used to ensure a generalizable result

by taking blocks that are heterogeneous, while eliminating block effects by ensuring that treatment comparisons are made within blocks. ■

Although described above only in the simplest cases, the normal linear model provides approximations to randomization distributions in other settings also. Below we continue to talk of normal errors, with the understanding that these often generate approximations to randomization distributions.

9.1.2 Causal inference

In many investigations the key question is causal. Does passive smoking cause lung cancer? Does exposure to air pollution increase levels of asthma? Does applying treatment T to a unit increase its response Y by amount δ ? The extensive philosophical discussion of causality is largely irrelevant here, because of its focus on deterministic relations between cause and effect. The best we can usually hope for is statements such as ‘if applied to a large sample of units, T would give an average increase δ , compared with what would have been observed had they remained untreated’. This translates into probability statements for individual units.

It is important to appreciate that potential causes are aspects of units that could in principle be manipulated in the context in question. In a study of the effects of lifestyle on longevity, we can conceive of altering individuals’ dietary and exercise habits, for example, but not their genders. We can imagine comparing the survival of flabby burger-loving Mr Jones with his survival as fit or vegetarian or both, but not with that of Mr Jones as female rather than male; were he a woman, he would not be Mr Jones. Here diet and exercise are potential causes, but gender is not. Intrinsic attributes of units cannot be regarded as potential causes, because to speak of a causal effect of T on Y , intervention to change the value of T must be possible.

Three types of causal statement are as follows. First, and strongest, there may be a well-understood evidence-based mechanism or set of mechanisms — biological, physical or whatever — that links a cause to its effect. This is the usual meaning of causality in so-called hard science, even though knowledge about the mechanism is invariably subject to improvement.

Second, and much weaker, is the observation that two phenomena are linked by a stable association, whose direction is established and which cannot be explained by mutual dependence on some other allowable variable. In Example 6.18, for example, ignoring age induced an apparently positive association between survival and smoking, whose direction was reversed once age was taken into account. To see this differently, consider a population of units on each of which (T, Y) may be observed, and let the association between T and Y be measured by

$$\gamma = E(Y \mid T = 1) - E(Y \mid T = 0) > 0,$$

say. This can be estimated by the difference in averages $\bar{Y}_1 - \bar{Y}_0$ for samples with $T = 1$ and $T = 0$. To say that the association cannot be explained away amounts to

asserting that no confounding variable X exists for which

$$\gamma(x) = E(Y \mid T = 1, X = x) - E(Y \mid T = 0, X = x) \equiv 0.$$

In practice this will need to be bolstered by careful study design, often considering together studies that account for different possible confounders. The restriction to allowable variables means amongst other things that X cannot itself be a response to T .

A third interpretation of causality, intermediate between the first two, is related to experimentation and relies on the notion of a *counterfactual*. Consider a unit, and let R_0 and R_1 represent its responses on setting $T = 0$ and $T = 1$; these three variables are assumed to have a joint distribution. If in fact $T = 0$, then R_0 is observed and R_1 is counterfactual; it is the response that would have been observed had the treatment been different. Conversely R_0 is counterfactual if $T = 1$. The central difficulty of causal inference is that it is impossible to compare values of R_0 and R_1 from the same unit. Thus an assumption of homogeneity of treatment effects over units, that is, unit-treatment additivity, is essential. If unit-treatment additivity holds then the effect of T is measured by the difference of mean responses

$$\delta = E(R_1) - E(R_0),$$

but unlike γ this is not observable. In general $\delta \neq \gamma$, but if the treatment allocation is randomized, then T is independent of any property of the unit, and the consistency equation $Y = R_0(1 - T) + R_1T$ relating the counterfactuals to the response Y entails

$$\begin{aligned} \delta &= E(R_1) - E(R_0) = E(R_1 \mid T = 1) - E(R_0 \mid T = 0) \\ &= E(Y \mid T = 1) - E(Y \mid T = 0) = \gamma. \end{aligned}$$

Hence unit-treatment additivity and randomization ensure that the quantity δ we want to estimate equals γ , which we can estimate.

This argument presumes there to be no relation between treatment allocation and any property of the unit, and this is typically true only in completely randomized experiments. Suppose however that (R_1, R_0) and T are independent conditional on the value of another variable X , as in the lower right of Figure 9.1. Then

$$E(R_1) = E_X\{E(R_1 \mid X)\} = E_X\{E(R_1 \mid X, T = 1)\} = E_X\{E(Y \mid X, T = 1)\},$$

and with a parallel argument for $E(R_0)$ we have

$$\delta = E(R_1) - E(R_0) = E_X\{E(Y \mid X, T = 1) - E(Y \mid X, T = 0)\} = \gamma,$$

say. The observable effect γ , a function of the joint distribution of Y , X , and T , is now averaged over the possible values of X for the unit. The interpretation of γ and the case for a causal effect are both strengthened if in fact

$$E(Y \mid X, T = 1) - E(Y \mid X, T = 0) = \gamma(X) \equiv \gamma,$$

that is, association with T does not depend on X , as in Figure 9.2. Otherwise there is *interaction* between T and Y ; see Section 9.3.1.

The assumption need not apply on the original scale; it might apply to a transformed response, in which case the argument below is applied on the transformed scale.

The use of randomization to eliminate confounding variables is a powerful tool, but it is not sufficient for causal inference. An obvious counter-example is the left panel of Figure 9.2, where it would be foolhardy to talk of a causal effect of T on Y even if the appropriate linear model had been fitted, because the observed triples (x, t, y) give no way to assess whether confounding between X and T is present despite randomization.

Even if experimentation has established that T changes the distribution of Y , it seems rash to assert causality with no idea of an underlying mechanism. In practice a combination of evidence from physical mechanisms, direct experiment, and large-scale observational data will be most compelling.

Exercises 9.1

- (a) Show that under the two-sample model, the difference of the sample averages, $\bar{y}_2 - \bar{y}_1$, has variance $(n_1 + n_2)\sigma^2/(n_1n_2)$. Show that subject to $n_1 + n_2 = n$, this is minimized when n_1 and n_2 are as nearly equal as possible.

(b) Suppose that n units are split into k blocks of size $m + 1$, and that one unit in each block is chosen at random to be treated, while the remaining m are controls. Suppose that the responses in the j th block are y_{j1} and $y_{j2}, \dots, y_{j(m+1)}$, and let d_j represent the difference between the treated individual and the average of the controls. Show that the average of these differences has variance $(m + 1)\sigma^2/(km)$, and show that for fixed n this is minimized when $m = 1$.
- Suppose a paired comparison experiment is performed, in which the j th pair satisfies the normal linear model

$$y_{0j} = \mu_j - \delta + \varepsilon_{0j}, \quad y_{1j} = \mu_j + \delta + \varepsilon_{1j}, \quad j = 1, \dots, m,$$

but that data analysis is performed using the two-sample model. Show that the variance estimator can be written as

$$S^2 = \frac{1}{2(m-1)} \sum_{j,t} (\mu_j - \bar{\mu} + \varepsilon_{tj} - \bar{\varepsilon}_{..})^2.$$

Deduce that this has expected value $\sigma^2 + (m-1)^{-1} \sum_j (\mu_j - \bar{\mu})^2$ conditional on the μ_j , and hence show that if the μ_j are normally distributed with variance τ^2 , then $E(S^2) = \sigma^2 + \tau^2$.

Show that if the two-sample model is used in this situation, the length of a 95% confidence interval for 2δ is roughly $2(\sigma^2 + \tau^2)^{1/2} t_{2(m-1)}(0.025)$, whereas under the paired comparisons model the length is about $2\sigma t_{m-1}(0.025)$. For what values of τ^2/σ^2 are the two-sample intervals shorter when (a) $m = 3$, (b) $m = 11$? Discuss your results.

- Check (9.3), find $\text{var}(T_j)$ and $\text{cov}(T_j, T_k)$ and hence verify the given formulae for the mean and variance of $\bar{Y}_1 - \bar{Y}_0$.
- In Example 9.1, show that Z is a monotonic function of $\bar{D}$.
- To what extent can gender be regarded as a cause in studies (a) relating longevity and lifestyle and (b) of salary differentials in employment?
- Let $T = 0$ with probability $1 - \alpha$ and $T = 1$ otherwise, and suppose that conditional on $T = 0$, R_0 is normal with mean zero and R_1 is normal with mean δ , while conditional on $T = 1$, the corresponding means are η and $\eta + \delta$; in each case the variables have unit variances. Let $Y = R_0(1 - T) + R_1T$ denote the observed response variable. Show that $\gamma = E(Y | T = 1) - E(Y | T = 0) = \eta + \delta$, and deduce that $\delta = E(R_1) - E(R_0)$ cannot be estimated unless (R_0, R_1) and T are independent.

Term	df	Sum of squares	Mean square
Groups	$T - 1$	$\sum_{t,r} (\bar{y}_{t.} - \bar{y}_{..})^2$	$(T - 1)^{-1} \sum_t (\bar{y}_{t.} - \bar{y}_{..})^2$
Residual	$T(R - 1)$	$\sum_{t,r} (y_{tr} - \bar{y}_{t.})^2$	$\{T(R - 1)\}^{-1} \sum_{t,r} (y_{tr} - \bar{y}_{t.})^2$

Table 9.2 Analysis of variance table for one-way layout.

9.2 Some Standard Designs

9.2.1 One-way layout

If more than two treatments are to be compared and the population is relatively homogeneous, the two-group model may be extended to a completely randomized design, known as a one-way layout. Henceforth we let T denote the number of treatments in the model under consideration.

Suppose that we wish to compare the effects of T treatments and that we have available $n = RT$ units. We divide the units at random into T groups each of size R , and apply a single treatment to all the units in each group. The corresponding linear model is

$$y_{tr} = \beta_t + \varepsilon_{tr}, \quad t = 1, \dots, T, \quad r = 1, \dots, R, \quad (9.4)$$

where $\varepsilon_{tr} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$. This assumes that the only effect of the treatment is to alter the mean response, as would be the case under a randomization distribution. Thus the observations within each group are random samples, but the groups may have different means. This explains the term one-way layout: laid out as a $T \times R$ array, only the treatment index is meaningful. In matrix terms this model is

$$\begin{pmatrix} y_{11} \\ \vdots \\ y_{1R} \\ y_{21} \\ \vdots \\ y_{2R} \\ \vdots \\ y_{T1} \\ \vdots \\ y_{TR} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_T \end{pmatrix} + \begin{pmatrix} \varepsilon_{11} \\ \vdots \\ \varepsilon_{1R} \\ \varepsilon_{21} \\ \vdots \\ \varepsilon_{2R} \\ \vdots \\ \varepsilon_{T1} \\ \vdots \\ \varepsilon_{TR} \end{pmatrix}. \quad (9.5)$$

This design matrix has full rank T and the least squares estimator of β_t it yields is the average for the t th group, $\bar{y}_{t.} = R^{-1} \sum_r y_{tr}$. If the β_t are all equal, corresponding to the model $y = 1_n \beta_0 + \varepsilon$ in our general notation, the fitted value for the entire set of data is the overall average $\bar{y}_{..}$. The sum of squares then decomposes as

$$\sum_{t,r} (y_{tr} - \bar{y}_{..})^2 = \sum_{t,r} (y_{tr} - \bar{y}_{t.})^2 + \sum_{t,r} (\bar{y}_{t.} - \bar{y}_{..})^2,$$

corresponding to (8.23), and the analysis of variance is shown in Table 9.2.

Here and below replacement of a subscript by a dot indicates averaging over the values of that subscript.

Table 9.3 Data on the teaching of arithmetic.

Group		Test result y								Average	Variance
A (Usual)	17	14	24	20	24	23	16	15	24	19.67	17.75
B (Usual)	21	23	13	19	13	19	20	21	16	18.33	12.75
C (Praised)	28	30	29	24	27	30	28	28	23	27.44	6.03
D (Reproved)	19	28	26	26	19	24	24	23	22	23.44	9.53
E (Ignored)	21	14	13	19	15	15	10	18	20	16.11	13.11

The unbiased estimator of σ^2 when (9.4) is fitted is

$$S^2 = \frac{1}{n-p} (y - \hat{y})^T (y - \hat{y}) = \frac{1}{T(R-1)} \sum_{t,r} (y_{tr} - \bar{y}_{t.})^2,$$

with $T(R-1)$ degrees of freedom. If $R=1$ it is impossible to estimate σ^2 , for then there is only one observation with which to estimate β_t , and $y_{tr} \equiv \bar{y}_{t.}$. Thus *replication* of the responses for each treatment is essential unless an external estimate of σ^2 is available, for example from another experiment. A further benefit of replication is the capacity to check model assumptions, as we shall see in Examples 9.2 and 9.6.

The F statistic for assessing significance of differences among treatments,

$$F = \frac{(T-1)^{-1} \sum_t (\bar{y}_{t.} - \bar{y}_{..})^2}{S^2} \sim F_{T-1, T(R-1)},$$

when $\beta_1 = \dots = \beta_T$. In applications interest generally focuses on estimation of particular differences among the β_t , however, rather than on testing for overall differences, this being merely an initial screening device.

Another possible linear model for the data is

$$y_{tr} = \alpha + \gamma_t + \varepsilon_{tr}, \quad t = 1, \dots, T, \quad r = 1, \dots, R,$$

in which the overall mean is represented by α , and γ_t represents the difference between the mean for treatment t and the overall mean. The design matrix for this model has $T+1$ columns, namely the T columns of the matrix in (9.5) and a column of ones, and has rank T : the $T+1$ parameters cannot be estimated from T groups. Although the T linear combinations $\alpha + \gamma_1, \dots, \alpha + \gamma_T$ corresponding to the group means are estimable, the $T+1$ parameters $\alpha, \gamma_1, \dots, \gamma_T$ are not.

Example 9.2 (Teaching methods data) In an investigation on the teaching of arithmetic, 45 pupils were divided at random into five groups of nine. Groups A and B were taught in separate classes by the usual method. Groups C, D, and E were taught together for a number of days. On each day C were praised publicly for their work, D were publicly reproved and E were ignored. At the end of the period all pupils took a standard test, with the results given in Table 9.3 and displayed in the left panel of Figure 9.4. Groups A and B seem to have performed similarly, but the other groups have responded differently to the regimes imposed, as we see from the averages and variances in the final columns of the table. If the only differences among groups were in their means, the group variances could be expected to be independently distributed

Term	df	Sum of squares	Mean square	F
Groups	4	722.67	180.67	15.3
Residual	40	473.33	11.83	

Table 9.4 Analysis of variance for data on the teaching of arithmetic.

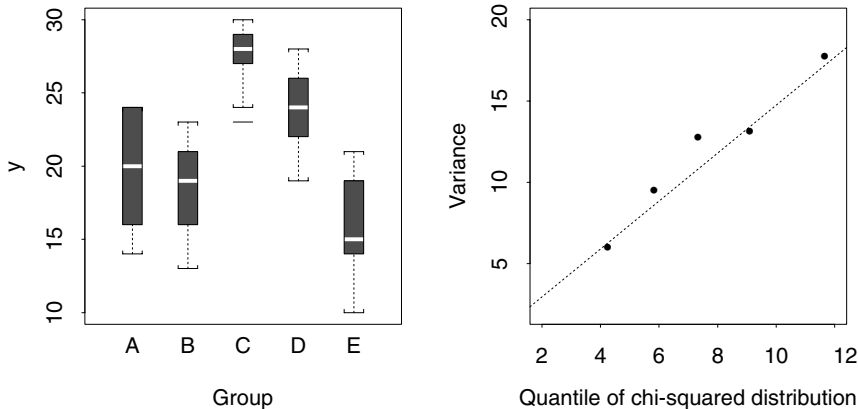


Figure 9.4 Data on teaching of arithmetic. The left panel shows the original data, and the right panel shows the ordered variances for each group plotted against plotting positions for the χ^2_8 distribution.

as $\sigma^2\chi^2_8/8$. No doubt is cast on this by the corresponding probability plot, shown in the right panel of Figure 9.4; this is only available because of the replication within each group.

The analysis of variance, shown in Table 9.4, shows very strong evidence of differences among the groups, as we would expect from inspecting the data. The corresponding F statistic is 15.3, to be considered as $F_{4,40}$ under the hypothesis of no group differences, in which case the significance level is zero.

As a group average is an average of $R = 9$ observations, its variance is σ^2/R , and consequently the estimated variance for the difference between the averages for groups A and B, $\bar{y}_A - \bar{y}_B = 1.33$, is $2s^2/9 = 2.63$. The corresponding t statistic, $1.33/2.63^{1/2}$, shows no evidence of differences between the control groups, and the pooled estimate of the mean using the usual teaching method, β_U , is accordingly $\bar{y}_U = \frac{1}{2}(19.67 + 18.33) = 19$, with estimated variance $s^2/18$.

Comparisons of the usual and other methods are of interest here, and they are based on statistics such as $\bar{Y}_C - \bar{Y}_U$, each having estimated variance $s^2/18 + s^2/9 = 1.97$. Confidence intervals for the underlying differences are based on the quantities $\{\bar{Y}_C - \bar{Y}_U - (\beta_C - \beta_U)\}/\{S^2/18 + S^2/9\}^{1/2}$, each having a t_{40} distribution. Thus 95% confidence intervals are (5.7, 11.2) for $\beta_C - \beta_U$, (1.7, 7.2) for $\beta_D - \beta_U$, and (−5.6, −0.14) for $\beta_E - \beta_U$. Giving approval and reproval improves test performance relative to the usual method, with approval working best, while ignoring pupils decreases their test scores, though by less. These conclusions are necessarily highly tentative because of the very limited scale of the experiment. ■

9.2.2 Randomized block design

Suppose that T treatments are to be compared, and that $n = TB$ units are available. The analogue of the paired comparisons experiment when there are more than two treatments is the *randomized block design*. The units are divided into B blocks of T units so that similar units are so far as possible in the same block. The T treatments are then applied randomly to the units, each treatment appearing precisely once in each block. A simple linear model here is that the response of the unit in block b given treatment t is

$$y_{tb} = \mu + \alpha_t + \beta_b + \varepsilon_{tb}, \quad t = 1, \dots, T, b = 1, \dots, B, \quad (9.6)$$

where the ε_{tb} are a random sample of $N(0, \sigma^2)$ variables. This is the *two-way layout* model, so-called because the y_{tb} can be laid out as an array with T rows and B columns, with α_t the treatment effect for the t th row and β_b the block effect for the b th column; see Table 9.6. With $T = 4$ and $B = 3$ for definiteness, and with parameter vector $(\mu, \alpha_1, \alpha_2, \alpha_3, \alpha_4, \beta_1, \beta_2, \beta_3)^T$, the 12×8 design matrix

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

has rank $1 + (T - 1) + (B - 1) = 6$: all eight parameters cannot be estimated. The terms corresponding to the treatment and block effects are columns 2–5 and 6–8 of this matrix respectively. Dropping the second and sixth columns of X is equivalent to setting $\alpha_1 = \beta_1 = 0$, in which case α_t and β_b represent the mean differences in response between treatment t and treatment 1 and between block b and block 1. In this, the *corner-point* parametrization, μ is the mean response of the unit in block 1 given treatment 1, that is, in the top left corner of the two-way layout. The least squares estimates in this parametrization can be obtained from the usual formula, but their derivation is unenlightening.

Instead, let us use the original parametrization with the least squares estimates constrained so that $\sum_r \hat{\alpha}_r = \sum_c \hat{\beta}_c = 0$. These are constraints on the estimates, not on the parameters: nature is free to use as many parameters as she likes, but only certain linear combinations of them are estimable. These two linear restrictions ensure that our fitted model is not overparametrized, and we can use symmetry to avoid inverting a rank-deficient matrix $X^T X$. We use Lagrange multipliers to find the values of $\hat{\mu}, \hat{\alpha}_t,$

$\hat{\beta}_b$, η and ζ that minimize

$$\sum_{t,b} (y_{tb} - \hat{\mu} - \hat{\alpha}_t - \hat{\beta}_b)^2 + \eta \left(\sum_t \hat{\alpha}_t - 0 \right) + \zeta \left(\sum_b \hat{\beta}_b - 0 \right).$$

On differentiating, we see that we should solve the equations

$$\begin{aligned} 0 &= \sum_{t,b} (y_{tb} - \hat{\mu} - \hat{\alpha}_t - \hat{\beta}_b), \\ 0 &= \sum_b (y_{tb} - \hat{\mu} - \hat{\alpha}_t - \hat{\beta}_b) - \eta, \quad t = 1, \dots, T, \\ 0 &= \sum_t (y_{tb} - \hat{\mu} - \hat{\alpha}_t - \hat{\beta}_b) - \zeta, \quad b = 1, \dots, B, \\ 0 &= \sum_t \hat{\alpha}_t, \quad 0 = \sum_b \hat{\beta}_b, \end{aligned}$$

giving $\hat{\mu} = \bar{y}_{..}$, $\hat{\alpha}_t = \bar{y}_{t.} - \bar{y}_{..}$, and $\hat{\beta}_b = \bar{y}_{.b} - \bar{y}_{..}$, where as before we use a dot in a subscript to indicate averaging over the corresponding index. Thus we have $\bar{y}_{..} = (TB)^{-1} \sum_{t,b} y_{tb}$, $\bar{y}_{t.} = B^{-1} \sum_b y_{tb}$, and so forth.

The fitted values are $\hat{\mu} + \hat{\alpha}_t + \hat{\beta}_b = \bar{y}_{.b} + \bar{y}_{t.} - \bar{y}_{..}$, and hence the residual sum of squares is $\sum_{t,b} (y_{tb} - \bar{y}_{t.} - \bar{y}_{.b} + \bar{y}_{..})^2$; these would be the same in the corner-point parametrization, because the same subspace is spanned by the columns of the design matrix in both cases, and the fitted values — though not the parameter estimates — depend only on the column space of the design matrix; recall Figure 8.2. If the $\hat{\beta}_b$ were not in the model, $\hat{\mu}$ and $\hat{\alpha}_t$ would remain the same, and the fitted values would be $\hat{\mu} + \hat{\alpha}_t = \bar{y}_{t.}$

The mean difference in response between treatments r and s is estimated by the difference $\hat{\alpha}_r - \hat{\alpha}_s = \bar{y}_{r.} - \bar{y}_{s.}$. If we write this estimate in terms of the underlying parameters, by replacing y_{rb} by $\mu + \alpha_r + \beta_b + \varepsilon_{rb}$ and so forth, we obtain

$$\begin{aligned} &B^{-1}(B\mu + B\alpha_r + \beta_1 + \dots + \beta_B + \varepsilon_{r1} + \dots + \varepsilon_{rB}) \\ &- B^{-1}(B\mu + B\alpha_s + \beta_1 + \dots + \beta_B + \varepsilon_{s1} + \dots + \varepsilon_{sB}), \end{aligned}$$

which equals $\alpha_r - \alpha_s + \bar{\varepsilon}_{r.} - \bar{\varepsilon}_{s.}$, and this is independent of the block effects, which appeared equally often in $\bar{y}_{r.}$ and $\bar{y}_{s.}$. Thus because the design is balanced, comparisons among treatments are essentially made within blocks, and this increases precision if there are substantial block effects.

The difference between an observation and the overall average equals

$$y_{tb} - \bar{y}_{..} = (y_{tb} - \bar{y}_{t.} - \bar{y}_{.b} + \bar{y}_{..}) + (\bar{y}_{t.} - \bar{y}_{..}) + (\bar{y}_{.b} - \bar{y}_{..}),$$

and because $\sum_{t,b} (y_{tb} - \bar{y}_{..})(\bar{y}_{t.} - \bar{y}_{..}) = 0$, $\sum_{t,b} (\bar{y}_{t.} - \bar{y}_{..})(\bar{y}_{.b} - \bar{y}_{..}) = 0$, and so forth, we see that

$$\sum_{t,b} (y_{tb} - \bar{y}_{..})^2 = \sum_{t,b} (y_{tb} - \bar{y}_{t.} - \bar{y}_{.b} + \bar{y}_{..})^2 + \sum_{t,b} (\bar{y}_{t.} - \bar{y}_{..})^2 + \sum_{t,b} (\bar{y}_{.b} - \bar{y}_{..})^2,$$

which echoes (8.23). If we had set $\hat{\beta}_b \equiv 0$, the corresponding sum of squares

Table 9.5 Analysis of variance table for two-way layout model.

Term	df	Sum of squares
Treatments	$T - 1$	$\sum_{t,b} (\bar{y}_{t.} - \bar{y}_{..})^2$
Blocks	$B - 1$	$\sum_{t,b} (\bar{y}_{.b} - \bar{y}_{..})^2$
Residual	$(T - 1)(B - 1)$	$\sum_{t,b} (y_{tb} - \bar{y}_{t.} - \bar{y}_{.b} + \bar{y}_{..})^2$

Table 9.6 Data on weight gains in pigs.

Diet	Group								Average
	1	2	3	4	5	6	7	8	
I	1.40	1.79	1.72	1.47	1.26	1.28	1.34	1.55	1.48
II	1.31	1.30	1.21	1.08	1.45	0.95	1.26	1.14	1.21
III	1.40	1.47	1.37	1.15	1.22	1.48	1.31	1.27	1.33
IV	1.96	1.77	1.62	1.76	1.88	1.50	1.60	1.49	1.70
Average	1.52	1.58	1.48	1.37	1.45	1.30	1.38	1.36	1.43

decomposition would have been

$$\sum_{t,b} (y_{tb} - \bar{y}_{..})^2 = \sum_{t,b} (y_{tb} - \bar{y}_{t.})^2 + \sum_{t,b} (\bar{y}_{t.} - \bar{y}_{..})^2,$$

and it follows by symmetry that once the constant term has been fitted, the reductions in sum of squares due to treatment and block terms are respectively $\sum_{t,b} (\bar{y}_{t.} - \bar{y}_{..})^2$ and $\sum_{t,b} (\bar{y}_{.b} - \bar{y}_{..})^2$. As $\sum_{t,b} (\bar{y}_{t.} - \bar{y}_{..})(\bar{y}_{.b} - \bar{y}_{..}) = 0$, these sums of squares are independent if the errors are normal.

The analysis of variance table for a two-way layout with T rows and B columns is in Table 9.5. The residual degrees of freedom are

$$TB - 1 - (T - 1) - (B - 1) = (T - 1)(B - 1),$$

and the sums of squares are independent of the order in which terms are fitted.

Example 9.3 (Pig diet data) Twelve pigs were divided into eight groups of four, in such a way that the pigs in any one group were expected to gain weight at equal rates if fed in the same way. Four diets were compared by randomly assigning them to pigs, subject to each diet occurring once in each group. The average daily weight gains of the pigs are given in Table 9.6. The diet averages suggest that pigs on diet IV gain more weight than the others, and that any differences between II and III are small. Differences among the groups are less marked.

The analysis of variance in Table 9.7 shows strong differences among diets, but little effect of blocking into groups. The estimate of σ^2 is $s^2 = 0.024$. The diet averages are 1.48, 1.21, 1.33, and 1.70, and as the standard error for a difference of two of them is $(2s^2/8)^{1/2} = 0.077$, it is clear that diet IV leads to the fastest weight gain, with diet I second and better than diet III; it is less clear that II is worse than III. ■

Term	df	Sum of squares	Mean square	F statistic
Diet	3	1.042	0.347	14.6
Group	7	0.247	0.035	1.48
Residual	21	0.500	0.024	

Table 9.7 Analysis of variance table for two-way layout model applied to the data of Table 9.6.

Embryo	Treat	y	Treat	y	Embryo	Treat	y	Treat	y
1	—	2.51	His—	2.15	9	His—	2.32	Lys—	2.53
2	—	2.49	Arg—	2.23	10	Arg—	2.15	Thr—	2.23
3	—	2.54	Thr—	2.26	11	Arg—	2.34	Val—	2.15
4	—	2.58	Val—	2.15	12	Arg—	2.30	Lys—	2.49
5	—	2.65	Lys—	2.41	13	Thr—	2.20	Val—	2.18
6	His—	2.11	Arg—	1.90	14	Thr—	2.26	Lys—	2.43
7	His—	2.28	Thr—	2.11	15	Val—	2.28	Lys—	2.56
8	His—	2.15	Val—	1.70					

Table 9.8 Log_{10} dry weight y (μg) of chick bones after cultivation over a nutrient chemical medium, either complete (—), or with single amino acids missing (Cox and Snell, 1981, p. 95). The order of treatment pairs was randomized, but the table shows them systematically.

Balanced incomplete block design

Sometimes variation among units is large enough for blocking to be required, but a randomized block design cannot be used because the block size is smaller than the number of treatments, T . In such circumstances it may be possible to use a *balanced incomplete block design*. Suppose that there are B blocks each with K units, and that $R = BK/T$ is an integer. In the simplest such design, each treatment appears exactly once in a block, and each pair of treatments appears together λ times, in which case $R(K - 1) = (T - 1)\lambda$.

Example 9.4 (Chick bone data) Table 9.8 gives data on the growth of chick bones. Bones from 7-day-old chick embryos were cultivated over a nutrient chemical medium. Two bones were available from each chick, and the experiment was set out in a balanced incomplete block design with two units per block. The treatments were growth in the complete medium, with about 30 nutrients in carefully controlled quantities, and growth in five other media, each with a single amino acid omitted. Thus His—, Arg—, and so forth denote media without particular amino acids. This balanced incomplete block design has $T = 6$, $B = 15$, $K = 2$, $R = 5$, and $\lambda = 1$.

One way to proceed here is to let $\beta_H, \beta_A, \dots$ denote the effect of the absence of His, Arg, $\dots$, and then regard the first pair of responses as having means $\mu_1, \mu_1 + \beta_H$, the sixth as having means $\mu_6 + \beta_H, \mu_6 + \beta_A$, and so forth. We then perform a linear regression with response the differences of the responses for each of the embryos, parameter vector $(\beta_H, \beta_A, \beta_T, \beta_V, \beta_L)^T$, and a 15×5 design matrix whose first and sixth rows are $(1, 0, 0, 0, 0)$ and $(-1, 1, 0, 0, 0)$. This avoids the necessity to estimate $\mu_1, \dots, \mu_{15}$, but gives the same estimates of the β s, shown in the first line of Table 9.9. The estimate of error variance is $s^2 = 0.013$. The initial sum of squares of 1.024 reduces to 0.132, giving overall F statistic 13.6 on 5 and 10 degrees of freedom: a

Table 9.9 Parameter estimates and standard errors for intra-block, inter-block and pooled analyses of chick data.

Analysis	Amino acid					SE
	His	Arg	Thy	Val	Lys	
Intra-block $\hat{\beta}$	-0.22	-0.35	-0.35	-0.49	-0.16	0.066
Inter-block $\hat{\beta}$	-0.55	-0.40	-0.33	-0.42	0.07	0.124
Pooled β^*	-0.29	-0.36	-0.34	-0.47	-0.11	0.058

highly significant reduction. Lack of each amino acid reduces growth; Lys has the smallest effect, but even this has the large t value of $-0.16/0.066 = -2.42$ on 10 degrees of freedom.

In this regression the terms for individual amino acids are not orthogonal, and the analysis of variance is not unique. For example, if acids are fitted in the order His, Arg, Thr, Val, Lys, the reductions in sum of squares are 0.014, 0.052, 0.078, 0.67, 0.077, while the reductions for the order Lys, Thr, Val, His, Arg are 0.074, 0.033, 0.40, 0.01, 0.38. Here balance gives equal precision for estimation of each of the β s, rather than orthogonal sums of squares.

Though simple, this so-called *intra-block* analysis uses a degree of freedom to estimate each block parameter μ_j , and if these vary little then information may be lost. To outline how *inter-block* analysis can retrieve this, we denote the responses from the j th block as y_{j1}, y_{j2} and treat these as independent normal variables with means $\mu_j + x_{j1}^T \beta$, $\mu_j + x_{j2}^T \beta$ and variances σ^2 . The previous analysis was based on $y_{j1} - y_{j2}$, and this is independent of the block sum $y_{j1} + y_{j2}$. The inter-block analysis treats the μ_j as random variables with mean μ , say, and variance σ_μ^2 , so the block sums have variance $2(2\sigma_\mu^2 + \sigma^2)$, perhaps much larger than the variance $2\sigma^2$ of the differences. We then fit to the 15 block sums a linear model with means $\mu 1_{15} + X\beta$, the j th row of X being $x_{j1}^T + x_{j2}^T$, thereby obtaining estimates $\hat{\beta}$ of the amino acid effects. These estimates are independent of those obtained from the intrablock analysis; their values are given in Table 9.9, along with the standard error inflated by σ_μ^2 . Both sets of estimates are unbiased, so approximate minimum variance unbiased estimates are formed as a weighted combination $\beta^* = w\hat{\beta} + (1-w)\hat{\beta}$, where $w = \hat{v}^{-1}/(\hat{v}^{-1} + \hat{v}^{-1})$, $\hat{v}$ and $\hat{v}$ being the estimated variances for the intra- and inter-block estimates. As $w = 0.78$, these pooled estimates are close to the $\hat{\beta}$, with a slightly smaller standard error. This standard error combines independent standard errors from the intra- and inter-block analyses and has approximately 13.8 degrees of freedom; see Exercise 9.2.3.

The response is $\log_{10}$ dry weight, so the effect of eliminating an amino acid is multiplicative on the original scale. A 0.95 confidence interval for the median effect of eliminating His is $10^{\hat{\beta}_H \pm 2.15 \times 0.058} = (0.38, 0.68)$, with the estimate $10^{\hat{\beta}_H} = 0.51$ corresponding to a 50% reduction in growth. See Practical 9.1. ■

$$t_{13.8}(0.975) = 2.15$$

There are many generalizations of incomplete block designs. Their key purpose is to give good precision for estimation of the effects of interest — usually treatment effects, or perhaps a subset of them — when there are more treatments than blocks.

Higher degrees of balance are possible, in which all triples of treatments appear equally often, and sometimes constraints on the numbers of units lead to the use of partially balanced designs, which give increased precision on treatments of primary interest while sacrificing precision on those of less importance.

9.2.3 Latin square

When there are two possible blocking factors, a three-way layout could be used. In many circumstances, however, a design that requires fewer units is required, and one possibility may be a *Latin square*. Suppose that the blocking factors and the treatment have the the same number of levels, q . Then a Latin square design is constructed by laying out units in a $q \times q$ array with the blocking factors corresponding to rows and columns, and applying each treatment precisely once in each row and in each column. An example is shown in the upper left part of Table 9.11. This balanced application of treatments leads to an orthogonal decomposition of the total sum of squares. Many such layouts are possible, with randomization by choice of design and permutation of row, column and treatment labels.

The corresponding linear model treats the response in the r th row and c th column as $y_{rc} = \mu + \alpha_r + \beta_c + \gamma_{t(r,c)} + \varepsilon_{rc}$, where $t(r, c)$ is the treatment applied to that unit, and $\varepsilon_{rc} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$. As it stands this model contains $1 + 3q$ parameters but the design matrix would have rank $1 + 3(q - 1)$.

Least squares estimates may be obtained by extending the Lagrange multiplier argument on page 429. We minimize $\sum_{r,c} (y_{rc} - \hat{\mu} - \hat{\alpha}_r - \hat{\beta}_c - \hat{\gamma}_{t(r,c)})^2$ subject to the constraints $\sum_r \hat{\alpha}_r = \sum_c \hat{\beta}_c = \sum_t \hat{\gamma}_t = 0$. This yields

$$\hat{\mu} = \bar{y}, \quad \hat{\alpha}_r = \bar{y}_{r\cdot} - \bar{y}_{\cdot\cdot}, \quad \hat{\beta}_c = \bar{y}_{\cdot c} - \bar{y}_{\cdot\cdot}, \quad \hat{\gamma}_t = \bar{y}_{t\cdot} - \bar{y}_{\cdot\cdot},$$

with residual sum of squares $\sum_{r,c} (y_{rc} - \bar{y}_{r\cdot} - \bar{y}_{\cdot c} - \bar{y}_{t\cdot} + 2\bar{y}_{\cdot\cdot})^2$.

To see this another way, suppose that we had ignored the treatment classification in the Latin square, and obtained the analysis of variance table for the row and column classifications. These would be the same as in Table 9.5, though the residual sum of squares would also contain the variation due to treatments. However, we could rewrite the table so that treatments appeared as the row classification, in which case the current row classification would take on the role of treatments and appear inside the table. The two-way analysis of variance table for the rearranged data, ignoring the new treatments (old rows), would contain sums of squares due to treatments and to columns, and its residual sum of squares would also contain the sum of squares for rows. Since the sums of squares for both two-way analyses are orthogonal, the analysis of variance table for a $q \times q$ Latin square must be as shown in Table 9.10.

Example 9.5 (Field concrete mixer data) A field concrete mixer lays down a concrete road surface while moving forward. Its efficiency is measured by the hardness of the surface it produces, as a percentage of the corresponding hardness produced under laboratory conditions. It is thought that efficiency may fall off as the speed at which the machine moves increases, and trials were performed to investigate this. On each of four days, the machine was run at four different speeds, 4, 8, 12, and

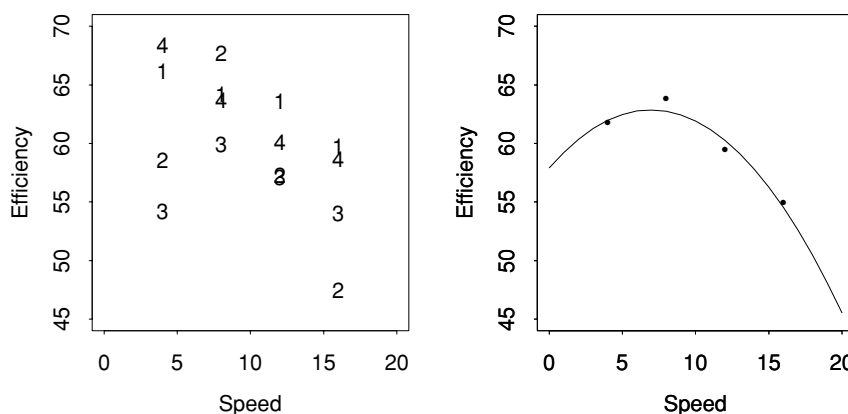
Table 9.10 Analysis of variance table for a Latin square.

Term	df	Sum of squares
Rows	$q - 1$	$\sum_{r,c} (\bar{y}_{r\cdot} - \bar{y}_{\cdot\cdot})^2$
Columns	$q - 1$	$\sum_{r,c} (\bar{y}_{\cdot c} - \bar{y}_{\cdot\cdot})^2$
Treatments	$q - 1$	$\sum_{r,c} (\bar{y}_{t(r,c)} - \bar{y}_{\cdot\cdot})^2$
Residual	$(q - 1)(q - 2)$	$\sum_{r,c} (y_{rc} - \bar{y}_{r\cdot} - \bar{y}_{\cdot c} - \bar{y}_{t(r,c)} + 2\bar{y}_{\cdot\cdot})^2$

Table 9.11 Field concrete mixer data. Latin square experiment, showing application of treatments — speed in miles per hour (left) — and observed responses — machine efficiency (%) (right) — for 16 combinations of day and run. Below are average efficiencies for day, run, and speed.

Day	Run				Day	Run			
	1	2	3	4		1	2	3	4
1	8	16	4	12	1	64.2	59.8	66.2	63.6
2	16	12	8	4	2	47.5	57.3	67.7	58.6
3	4	8	12	16	3	54.2	59.9	57.1	54.1
4	12	4	16	8	4	60.1	68.4	58.7	63.7

Day	Average	Run	Average	Speed	Average
1	63.45	1	56.50	4	61.85
2	57.78	2	61.35	8	63.88
3	56.33	3	62.43	12	59.53
4	62.73	4	60.00	16	55.03

Figure 9.5 Field concrete mixer data. Left panel: efficiencies as a function of speed, with plotting symbol giving the day. Right panel: average efficiencies, with fitted quadratic curve corresponding to day 1 and run 1.

16 miles per hour, these being taken in a different order on each day. The layout in Table 9.11 gives the speeds and the response, machine efficiency. There are two blocking factors, day and run, and a quantitative treatment with four levels, speed. The averages, also in Table 9.11, show large differences among days and speeds, and smaller ones among runs, while they and the left panel of Figure 9.5 show a systematic variation of efficiency with speed.

Term	df	Sum of squares	Mean square	F statistic
Days	3	151.06	50.35	5.78
Runs	3	79.74	26.58	3.05
Speeds	3	173.58	57.86	6.64
Residual	6	52.23	8.71	

Table 9.12 Analysis of variance for Latin square fitted to field concrete mixer data.

The analysis of variance in Table 9.12 shows evidence of day and speed differences, with significance levels for their F statistics respectively 0.03 and 0.02, and weak evidence of differences among runs, at significance level 0.11.

The estimated mean response for the t th treatment is $\hat{\mu} + \hat{\gamma}_t = \bar{y}_t$, and the average efficiencies for the speeds are 61.85, 63.88, 59.53, and 55.03%, suggesting that the best speed is 8 mph or so. A 95% confidence interval for the true efficiency at 8 mph is $63.88 \pm t_6(0.025)(s^2/4)^{1/2}$, and since $s^2 = 8.71$ and $t_6(0.025) = -2.45$, this interval is (60.26, 67.50)%. We return to these data in Example 9.12. ■

9.2.4 Factorial design

A factorial design involves a number of treatments, each with several levels, and every combination of levels of the different factors appears together. Thus if there are two factors with two levels and one with three levels, there are $2 \times 2 \times 3 = 12$ possible treatment combinations, each of which is applied to at least one unit. A 2^3 factorial design was used in Example 8.4.

Example 9.6 (Poisons data) The data in Table 8.10 are from a 3×4 factorial experiment with four replicates. The model

$$y_{tpj} = \mu + \alpha_t + \beta_p + \gamma_{tp} + \varepsilon_{tpj}, \quad t = 1, 2, 3, 4, \quad p = 1, 2, 3, \quad j = 1, 2, 3, 4, \quad (9.7)$$

modifies (8.28) by adding terms γ_{tp} representing the *interaction* of poisons and treatments; see Section 9.3.1. If the γ_{tp} are all equal, (9.7) is the two-way layout model (9.6), except that there are four replicates at each combination of the two factors. The model (9.7) has 20 parameters, which evidently cannot be estimated separately from the 12 groups of times available. If we use Lagrange multipliers to minimize the sum of squares subject to the constraints

$$\sum_t \hat{\alpha}_t = \sum_p \hat{\beta}_p = \sum_t \hat{\gamma}_{tp} = \sum_p \hat{\gamma}_{tp} = 0,$$

then we find

$$\hat{\mu} = \bar{y}_{...}, \quad \hat{\alpha}_t = \bar{y}_{t..} - \bar{y}_{...}, \quad \hat{\beta}_p = \bar{y}_{.p.} - \bar{y}_{...}, \quad \hat{\gamma}_{tp} = \bar{y}_{tp.} - \bar{y}_{t..} - \bar{y}_{.p.} + \bar{y}_{...},$$

with corresponding orthogonal decomposition

$$y_{tpj} - \bar{y}_{...} = (y_{tpj} - \bar{y}_{tp.}) + (\bar{y}_{tp.} - \bar{y}_{t..} - \bar{y}_{.p.} + \bar{y}_{...}) + (\bar{y}_{t..} - \bar{y}_{...}) + (\bar{y}_{.p.} - \bar{y}_{...}).$$

Table 9.13 Sums of squares for two-way layout with replication, assuming T rows, P columns, and J replicates in each cell. For the poisons data, $T = 4$, $P = 3$, and $J = 4$.

Term	df	Sum of squares
Rows	$T - 1$	$\sum_{t,p,j} (\bar{y}_{t..} - \bar{y}_{...})^2$
Columns	$P - 1$	$\sum_{t,p,j} (\bar{y}_{.p.} - \bar{y}_{...})^2$
Rows $\times$ Columns	$(T - 1)(P - 1)$	$\sum_{t,p,j} (\bar{y}_{tp.} - \bar{y}_{t..} - \bar{y}_{.p.} + \bar{y}_{...})^2$
Residual	$TP(J - 1)$	$\sum_{t,p,j} (y_{tpj} - \bar{y}_{tp.})^2$

Table 9.14 Analyses of variance for the poisons data, with responses y and y^{-1} . For MS and F read 'Mean square' and ' F statistic'.

Term	df	Response y			Response y^{-1}		
		SS	MS	F	SS	MS	F
Poisons	2	1.033	0.517	23.22	34.88	17.44	72.63
Treatments	3	0.921	0.307	13.81	20.41	6.80	28.34
Treatments $\times$ Poisons	6	0.250	0.042	1.87	1.57	0.26	1.09
Residual	36	0.801	0.022		8.64	0.24	

The sums of squares and their degrees of freedom are given in Table 9.13. Notice that if $J = 1$, the residual sum of squares is zero, because $y_{tpj} \equiv \bar{y}_{tp.}$, and the analysis of variance reduces to that in Table 9.5, with the interaction sum of squares used to estimate the error variance σ^2 . If it was known *a priori* that an interaction was likely to be present, replication would be essential rather than merely desirable, and if replication was for some reason impossible, an external estimate of σ^2 would be required.

The left part of Table 9.14 shows the analysis of variance. There are the expected strong effects of poisons and treatments, but the interaction is less important, with a significance level of about 0.11 when treated as an $F_{6,36}$ variable. The estimate of σ^2 is $s^2 = 0.022$.

Under the model (9.7) the fitted values for each cell are $\bar{y}_{tp.}$. This suggests a check on the adequacy of the model. The four observations within each combination of poison and treatment should form a random sample from the normal distribution with mean $\mu + \alpha_t + \beta_p + \gamma_{tp}$ and variance σ^2 , and therefore their sample variance s_{tp}^2 should have the $\sigma^2 \chi_3^2/3$ distribution if the error assumption is correct. The lower left panel of Figure 8.5 shows a systematic departure from linearity, suggesting that the assumption of normal errors with constant variance is untenable. The lower right panel shows that the inverse survival times follow the normal error model more closely, suggesting that it is better to replace y_{tpj} with y_{tpj}^{-1} . The corresponding analysis of variance, shown in the right part of Table 9.14, shows that the poison and treatment effects explain a higher proportion of the response variability on the inverse scale, and that the interaction is reduced.

With response y^{-1} , the parameter estimates for the model with no interaction are $\hat{\mu} = 2.62$, $\hat{\alpha}_1 = 0.90$, $\hat{\alpha}_2 = -0.76$, $\hat{\alpha}_3 = 0.32$, $\hat{\alpha}_4 = -0.46$, $\hat{\beta}_1 = -0.82$,

$\hat{\beta}_2 = -0.35$, $\hat{\beta}_3 = 1.17$. The standard errors are 0.07 for $\hat{\mu}$, 0.12 for the $\hat{\alpha}$ s, and 0.10 for the $\hat{\beta}$ s, all with units of (10-hours)⁻¹. As suggested by the panels of Figure 8.5, treatments B and D prolong life best, and poison 3 shortens it most. We reconsider these data in Example 9.8. ■

Exercises 9.2

- 1 Consider the one-way layout. Show that when the model $y_{tr} = \mu + \varepsilon_{tr}$ is fitted, the residual sum of squares is $\sum_{r,t} (y_{tr} - \bar{y}_{..})^2$ on $TR - 1$ degrees of freedom, where $\bar{y}_{..}$ is the overall average. Show that

$$\sum_{r,t} (y_{tr} - \bar{y}_{..})^2 = \sum_{r,t} (y_{tr} - \bar{y}_{t.})^2 + \sum_{r,t} (\bar{y}_{t.} - \bar{y}_{..})^2,$$

and hence verify the contents of Table 9.2.

How would you form a confidence interval for $\beta_1 - \beta_2$?

- 2 Calculate the analysis of variance table for the data of Example 9.3, and test whether there are differences between the diets and the groups. Find the standard error of a difference between diets, and use it to give a 95% confidence interval for the mean difference in weight gain between diets IV and I.
- 3 Suppose that T_1 and T_2 have common mean μ and variances σ_1^2 and σ_2^2 , and let S_1^2 and S_2^2 be estimators of σ_1^2 , σ_2^2 , independently distributed as $\chi_{\nu_1}^2$, $\chi_{\nu_2}^2$.
(a) Show that if σ_1^2 , σ_2^2 are known, then μ has minimum variance unbiased estimator

$$T = \frac{T_1/\sigma_1^2 + T_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2},$$

with variance $\sigma_1^2\sigma_2^2/(\sigma_1^2 + \sigma_2^2)$.

(b) Suppose that $\text{var}(T)$ is estimated by V in which σ_1^2 and σ_2^2 are replaced by their estimates. Show that V has approximate mean and variance

$$\frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}, \quad \frac{2\sigma_1^4\sigma_2^4}{(\sigma_1^2 + \sigma_2^2)^4} \left(\frac{\sigma_1^4}{\nu_1} + \frac{\sigma_2^4}{\nu_2} \right).$$

Hence show that if V is regarded as approximately χ^2 , then its degrees of freedom are $(\sigma_1^2 + \sigma_2^2)^2/(\sigma_1^4/\nu_1 + \sigma_2^4/\nu_2)$.

(c) Compute the degrees of freedom for this approximation in Example 9.4.

- 4 Give the analysis of variance table for a two-way layout with replication, when the numbers of replicates in the t th row and p th column, J_{tp} , are unequal.
- 5 Use Lagrange multipliers to verify the formulae for the estimates and fitted values given in Example 9.5, and hence check the contents of the analysis of variance table for a Latin square.
- 6 In Example 9.5, suppose that a confidence interval is required for the difference of the mean efficiencies between 8 and 4 mph. Show that owing to the balance of the experiment, a point estimate of this is just the difference between the average efficiencies for these speeds, and that its variance is $\frac{1}{2}\sigma^2$. Give the estimate of σ^2 ignoring day and run effects, that is, treating the data as a one-way layout with the four levels of speed as groups. How much longer is the corresponding confidence interval than when day and run effects are taken into account?

9.3 Further Notions

9.3.1 Interaction

Terms that do not act additively in a linear model are said to *interact*. An easy way to understand this is by example.

Example 9.7 (2^2 factorial experiment) A 2^2 factorial experiment involves a response measured at each combination of two factors each with two levels. As an illustration, consider an experiment to assess the effects of two fertilizers, in which the factors are addition or not of potash and nitrogen. If the cell means are

	No potash	Potash
No nitrogen	μ	$\mu + \beta$
Nitrogen	$\mu + \alpha$	$\mu + \alpha + \beta + \gamma$

and $\gamma = 0$, the effects act additively because the addition of potash increases the mean response by β whether or not nitrogen is present. Similarly the effect of nitrogen does not depend on the presence of potash.

The two treatments interact if γ is non-zero, because the effect of both treatments together is not the sum of the effects of adding them separately. The difference between the effects of adding potash when there is no nitrogen present and when there is nitrogen present is

$$\{(\mu + \alpha + \beta + \gamma) - (\mu + \alpha)\} - \{(\mu + \beta) - \mu\} = \gamma,$$

so we can view a non-zero interaction of the two fertilizers as a differential effect of adding potash depending on the presence or not of nitrogen. The average effect of adding potash, taken over both rows of the table, is then

$$\frac{1}{2}\{(\mu + \alpha + \beta + \gamma) - (\mu + \alpha) + (\mu + \beta) - \mu\} = \beta + \frac{1}{2}\gamma.$$

Thus if there is no interaction, β represents the average effect of adding potash whatever the level of nitrogen, but it loses this interpretation if γ is non-zero.

If the model is reparametrized to have cell means

	No potash	Potash
No nitrogen	$\beta_0 - \beta_1 - \beta_2 + \beta_3$	$\beta_0 - \beta_1 + \beta_2 - \beta_3$
Nitrogen	$\beta_0 + \beta_1 - \beta_2 - \beta_3$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

the overall mean is β_0 , the average effect of adding potash is

$$\frac{1}{2}\{(\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_1 - \beta_2 - \beta_3) + (\beta_0 - \beta_1 + \beta_2 - \beta_3) - (\beta_0 - \beta_1 - \beta_2 + \beta_3)\} = 2\beta_2,$$

and likewise the average effect of adding nitrogen is $2\beta_1$. The difference between the effects of adding potash when there is no nitrogen present and when it is present is

$$\frac{1}{2}\{[(\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_1 - \beta_2 - \beta_3)] - [(\beta_0 - \beta_1 + \beta_2 - \beta_3) - (\beta_0 - \beta_1 - \beta_2 + \beta_3)]\} = 2\beta_3.$$

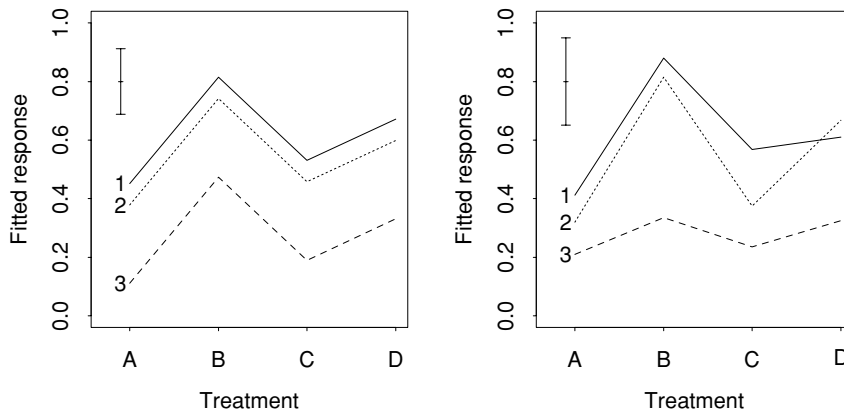


Figure 9.6 Poison data. The left panel shows how the fitted values under the model of no interaction, $\hat{\mu} + \hat{\alpha}_t + \hat{\beta}_p$, for treatments A–D depend on poisons 1–3. The right panel shows the corresponding fitted values under the model of interaction, $\hat{\mu} + \hat{\alpha}_t + \hat{\beta}_p + \hat{\gamma}_{tp}$. The vertical line in each panel has length four times the standard error of a fitted value.

The difference between the two parametrizations is clear from the design matrices,

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

The first parametrization, in terms of μ , α , β and γ , represents changes in the mean response relative to the top left cell; this is the *corner-point* parametrization. In the parametrization using β_0 , β_1 , β_2 and β_3 , the parameters can be interpreted as the overall mean, the mean effects of adding potash, nitrogen, and the effect of adding both, regardless of the other parameters. The first column corresponds to the overall mean, the middle two columns to *main effects* of potash and nitrogen, and the final column to the *first-order* or *two-factor* interaction between the two main effects. Notice that the interaction term is the product of the columns for the main effects and that the second parametrization is orthogonal, but the first is not. In practice the parametrization used would depend on the purpose of the analysis.

If interaction is present, four parameters must be estimated from four observations. In this case σ^2 would usually be estimated by replicating the experiment. ■

The same ideas generalize, as the next example shows.

Example 9.8 (Poisons data) The linear model corresponding to the data discussed in Example 9.6 is given at (9.7). If the first-order interaction parameters $\gamma_{tp} \equiv 0$, the profile of fitted values for poison p and treatments A–D may be written $(\hat{\mu} + \hat{\alpha}_1 + \hat{\beta}_p, \dots, \hat{\mu} + \hat{\alpha}_4 + \hat{\beta}_p)$, and the effect of applying poison r instead of poison p is a translation of the fitted values by $\hat{\beta}_r - \hat{\beta}_p$. The left panel of Figure 9.6 shows the profile of fitted values for the three poisons; of course the profiles are parallel, as would also be the case for the poison profiles $(\hat{\mu} + \hat{\alpha}_t + \hat{\beta}_1, \dots, \hat{\mu} + \hat{\alpha}_t + \hat{\beta}_3)$, because no interaction has been fitted.

Under the model with interaction, that is, including the γ_{tp} , the fitted values are $\hat{\mu} + \hat{\alpha}_t + \hat{\beta}_p + \hat{\gamma}_{tp} = \bar{y}_{tp}$, and the profiles are $(\bar{y}_{1p}, \dots, \bar{y}_{4p})$; these are shown in the right panel of the figure, and are not parallel, because under the model with interaction

The order of an interaction is one less than the number of effects it involves.

Table 9.15 Interactions for 2^3 factorial design.

Unit	Treatment	Intercept <i>I</i>	Main effects			Two-factor interactions			Three-factor interaction
			<i>A</i>	<i>B</i>	<i>C</i>	<i>AB</i>	<i>AC</i>	<i>BC</i>	<i>ABC</i>
1	1	+	−	−	−	+	+	+	−
2	<i>a</i>	+	+	−	−	−	−	+	+
3	<i>b</i>	+	−	+	−	−	+	−	+
4	<i>ab</i>	+	+	+	−	+	−	−	−
5	<i>c</i>	+	−	−	+	+	−	−	+
6	<i>ac</i>	+	+	−	+	−	+	−	−
7	<i>bc</i>	+	−	+	+	−	−	+	−
8	<i>abc</i>	+	+	+	+	+	+	+	+

the effect of changing poisons is more complex than a simple translation. The profiles are broadly similar to those in the left panel, and the evidence for interaction is weak, as shown by the F statistic for Treatments $\times$ Poisons in Table 9.14, which gives an overall test for departures from the simple pattern in the left panel. ■

Two-factor interactions are relatively common in applications. Higher-order interactions are rarer and can indicate outliers or a poorly fitting model.

Example 9.9 (2^3 factorial experiment) A 2^3 factorial experiment has three factors, A , B , and C , each with two levels, denoted -1 and $+1$. Each of the 2^3 possible combinations of the factor levels is applied to a unit, as in the main effects columns of Table 9.15, where the signs only are given. This design, replicated twice, is used in Example 8.4.

The second column of the table shows which treatments have been applied to each unit. Under the model with main effects of A , B , and C only, no treatment is applied to unit 1 and its mean response is $\beta_0 - \beta_A - \beta_B - \beta_C$, treatment A alone is applied to unit 2 and its mean is $\beta_0 + \beta_A - \beta_B - \beta_C$, and so forth. The design matrix then corresponds to the intercept and main effects columns, and

$$\hat{\beta}_A = \frac{1}{8}(y_a - y_1 + y_{ab} - y_b + y_{ac} - y_c + y_{abc} - y_{bc})$$

where y_a is the response for unit 2, y_{ab} is the response for unit 4, and so forth. Thus the estimate of β_A is based on contrasting the responses for units to which A was applied with those to which it was not applied. Likewise

$$\begin{aligned}\hat{\beta}_B &= \frac{1}{8}(y_b - y_1 + y_{ab} - y_a + y_{bc} - y_c + y_{abc} - y_{ac}), \\ \hat{\beta}_C &= \frac{1}{8}(y_c - y_1 + y_{ac} - y_a + y_{bc} - y_b + y_{abc} - y_{ab}).\end{aligned}$$

Under the model that includes the two-factor interactions β_{AB} , β_{AC} , and β_{BC} as well as the intercept and main effects, the mean response for unit 1 is $\beta_0 - \beta_A - \beta_B - \beta_C + \beta_{AB} + \beta_{AC} + \beta_{BC}$. On including the two-factor interaction columns in

the design matrix, we obtain

$$\hat{\beta}_{AB} = \frac{1}{8}(y_{ab} - y_a - y_b + y_1 + y_{abc} - y_{ac} - y_{bc} + y_c),$$

which is based on contrasting responses for which the levels of A and B are the same with those for which the levels of A and B are different. There are similar interpretations of the other estimated two-factor interactions

$$\hat{\beta}_{AC} = \frac{1}{8}(y_{ac} - y_a - y_c + y_1 + y_{abc} - y_{ab} - y_{bc} + y_b),$$

$$\hat{\beta}_{BC} = \frac{1}{8}(y_{bc} - y_b - y_c + y_1 + y_{abc} - y_{ab} - y_{ac} + y_a).$$

The estimated three-factor interaction is

$$\begin{aligned}\hat{\beta}_{ABC} &= \frac{1}{8}(y_{abc} + y_a + y_b + y_c - y_{ab} - y_{ac} - y_{bc} - y_1) \\ &= \frac{1}{8}\{(y_{abc} - y_{ac} - y_{bc} + y_c) - (y_{ab} - y_a - y_b + y_1)\},\end{aligned}$$

which contrasts the contributions to the AB interaction when C is applied and when it is not applied. Whichever of these models is fitted, the design matrix is orthogonal, and $(X^T X)^{-1} = \frac{1}{8}I$, so the variance of each of the estimates above is $\sigma^2/8$, as is readily verified directly. ■

When strong two-factor interaction is present, interpretation is often simplified by considering how responses behave separately for each level of one factor, and likewise for higher-order interactions.

Confounding

Factorial designs make it possible to assess the effects of many treatments and their interactions in a single experiment, but when there are many factors many homogeneous units must be found, and this can pose practical problems.

Example 9.10 (2^2 factorial experiment) An experiment with two two-level factors A and B is performed using two blocks each having two units. The possible treatments are 1, a , b , and ab , and there are three possible designs depending on which treatment appears in the same block as 1. Suppose that the first block has treatments 1 and a , and the second has b and ab . The model with an intercept, a block effect, and the main effects of A and B is

$$\begin{pmatrix} y_1 \\ y_a \\ y_b \\ y_{ab} \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 & -1 \\ 1 & 0 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \alpha \\ \beta_A \\ \beta_B \end{pmatrix} + \varepsilon,$$

in which the design matrix has rank three because the column for B is a linear combination of the first two columns. The design makes it impossible to distinguish these effects, which are said to be *confounded*. Evidently A would be confounded with blocks if the first block contained treatments 1 and b and the second a and ab .

Suppose instead that the experiment is set up to have 1 and ab in the same block. Then the design matrix is

$$\begin{pmatrix} 1 & 0 & -1 & -1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix},$$

which has rank four. Then $\hat{\beta}_0 = \frac{1}{2}(y_1 + y_{ab})$, $\hat{\alpha} = \frac{1}{2}(y_a + y_b - y_1 - y_{ab})$, while the estimates $\hat{\beta}_A = \frac{1}{4}(y_{ab} - y_1 + y_A - y_B)$ and $\hat{\beta}_B = \frac{1}{4}(y_{ab} - y_1 - y_A + y_B)$ correspond to comparisons made within blocks. Thus this design does allow the estimation of the main effects of interest, though an external estimate of the error variance σ^2 is required.

The interaction between A and B would usually be estimated by

$$\hat{\beta}_{AB} = \frac{1}{4}(y_a + y_b - y_1 - y_{ab}) = \frac{1}{2}\hat{\alpha},$$

so use of this design entails sacrificing any information about this interaction. In examples with many factors, interactions known to be unimportant are often deliberately confounded with blocks in such a way that the effects of interest are estimated with maximum precision in the resulting fractional factorial design. ■

If effects are confounded by accident, then further experimentation will be needed to identify the parameters, unless there is external information about their values. Some models are intrinsically non-identifiable, however; see Section 4.6.

9.3.2 Contrasts

Suppose that a model has $n \times p$ design matrix X of full rank and that we have a $p \times p$ invertible matrix A for which $XA = C$, where the first column of C is a column of ones, and the remaining columns, $c_1, \dots, c_{p-1}$, are orthogonal to the first and to each other. In this case $C^T C = \text{diag}(n, c_1^T c_1, \dots, c_{p-1}^T c_{p-1})$. Let us reparametrize the original model, $y = X\beta + \varepsilon$, by letting $\gamma = A^{-1}\beta$, thereby obtaining $y = C\gamma + \varepsilon$. The columns of C are known as *orthogonal contrasts*.

The least squares estimators for the model $y = C\gamma + \varepsilon$ are $\hat{\gamma} = (C^T C)^{-1} C^T y$, with covariance matrix $\sigma^2 (C^T C)^{-1}$. As $C^T C$ is a diagonal matrix, the estimate of γ_r is $\hat{\gamma}_r = c_r^T y / c_r^T c_r$, with variance $\text{var}(\hat{\gamma}_r) = \sigma^2 / c_r^T c_r$, and different estimates $\hat{\gamma}_r$ are uncorrelated with each other and with the overall average, $\bar{y}$. The residual sum of squares $SS(\hat{\gamma})$ equals

$$\begin{aligned} y^T \{I - C(C^T C)^{-1} C^T\} y &= y^T y - \hat{\gamma}^T C^T C \hat{\gamma} \\ &= \sum_{j=1}^n y_j^2 - n\bar{y}^2 - \hat{\gamma}_1^2 c_1^T c_1 - \dots - \hat{\gamma}_{p-1}^2 c_{p-1}^T c_{p-1}. \end{aligned}$$

As the reduction in sum of squares due to adding c_r to the design matrix is $\hat{\gamma}_r^2 c_r^T c_r$, the total sum of squares can be split into the contributions from each of the columns of

Term	df	Sum of squares	Mean square	F statistic
Seat	1	473.06	473.06	112.9
Dynamo	1	39.06	39.06	9.32
Tyre	1	39.06	39.06	9.32
Seat × Dynamo	1	1.56	1.56	0.37
Seat × Tyre	1	5.06	5.06	1.21
Dynamo × Tyre	1	0.06	0.06	0.01
Seat × Dynamo × Tyre	1	3.06	3.06	0.73
Residual	8	33.50	4.19	

Table 9.16 Analysis of variance for the cycling data.

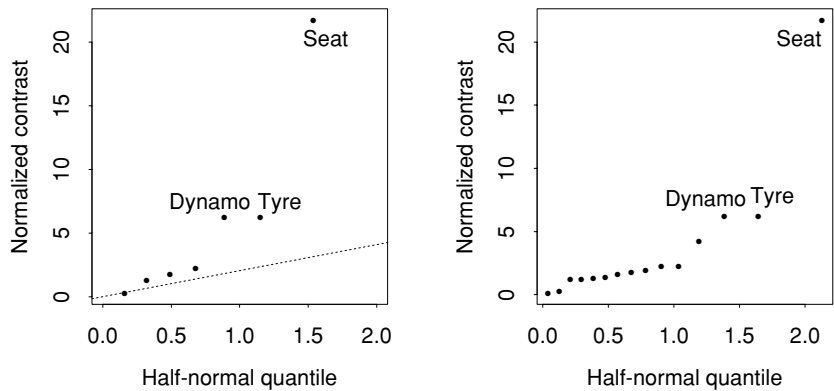


Figure 9.7 Half-normal plots of normalized contrasts $|\hat{\gamma}_r|(c_r^T c_r)^{1/2}$ for the data on cycling up a hill. The left panel shows the $|\hat{\gamma}_r|(c_r^T c_r)^{1/2}$ for the last seven columns of C_1 , with the dotted line having slope $s = 4.19^{1/2}$, the residual standard error. The right panel shows the $|\hat{\gamma}_r|(c_r^T c_r)^{1/2}$ for the last 15 columns of C_2 . In each case, only contrasts corresponding to main effects seem to be non-null. See text for details.

C , namely $n\bar{y}^2$, $\hat{\gamma}_1^2 c_1^T c_1$, and so forth. If γ_r equals zero, $\hat{\gamma}_r$ has mean zero and variance $\sigma^2/c_r^T c_r$, and if the errors are normal, $\hat{\gamma}_r^2 c_r^T c_r \sim \sigma^2 \chi_1^2$. A normal scores plot of the $\hat{\gamma}_r(c_r^T c_r)^{1/2}$, a plot of the ordered $\hat{\gamma}_r^2 c_r^T c_r$ against χ_1^2 plotting positions, or a half-normal plot (Practical 2.1) of the $|\hat{\gamma}_r|(c_r^T c_r)^{1/2}$, helps to show which of the γ_r may be non-zero.

Example 9.11 (Cycling data) The data on cycling up a hill are from a 2^3 factorial experiment, replicated twice. The design matrix D for a 2^3 experiment, in which the three main effects, the three second-order interactions, and the third-order interaction are all fitted, is obtained by adding ones to the pluses and minuses in Table 9.15. This matrix has the property that $D^T D = 8I_8$, so its columns are orthogonal contrasts. The 16×8 design matrix for the replicated experiment may be written as

$$C_1 = \begin{pmatrix} D \\ D \end{pmatrix};$$

as $C_1^T C_1 = 16I_8$, the columns of C_1 are also orthogonal contrasts. Table 9.16 shows the analysis of variance when this model is fitted. There are eight residual degrees of freedom, and the estimate of error is $s^2 = 4.19$. The main effects are significant, but the interactions, denoted $\text{Seat} \times \text{Dynamo}$ and so forth, are not.

The left panel of Figure 9.7 shows a half-normal plot of the quantities $|\hat{\gamma}_r|(c_r^T c_r)^{1/2}$ corresponding to the contrasts in the last seven columns of C_1 ; the dotted line has slope s . The plot confirms our impression from the analysis of variance table: only the three main effects seem to be non-zero.

The residual sum of squares can also be decomposed into its component degrees of freedom. To see how, we add eight columns to C_1 , giving

$$C_2 = \begin{pmatrix} D & -D \\ D & D \end{pmatrix},$$

the last 15 columns of which are orthogonal contrasts, as $C_2^T C_2 = 16I_{16}$. The right panel of Figure 9.7 shows a half-normal plot of the contrasts corresponding to these columns; the eight contrasts comprising s^2 — corresponding to the columns of C_2 not in C_1 — have been added to the previous seven. These eight contrast the main effects, the second- and third-order interactions between the two replicates. As no degrees of freedom remain with which to estimate σ^2 , it may be estimated by pooling those contrasts that lie roughly on a straight line in the lower left corner of the graph. Here there seem to be about 12 such contrasts, the pooling of which gives error estimate 3.60 on 12 degrees of freedom. ■

Example 9.12 (Field concrete mixer data) In Example 9.5 we found evidence that the efficiency of the mixer depended strongly on its speed, with the best concrete produced at about 8 mph. An estimated best speed can be obtained by decomposing the sum of squares due to speed using contrasts based on orthogonal polynomials. There are three degrees of freedom for speeds. One parametrization for them gives three columns in which the rows corresponding to the four speeds 4, 8, 12, and 16 miles per hour are

$$\begin{array}{cccc} 4 & 0 & 0 & 0 \\ 8 & 1 & 0 & 0 \\ 12 & 0 & 1 & 0 \\ 16 & 0 & 0 & 1 \end{array},$$

whereas a parametrization in terms of orthogonal polynomials gives

$$\begin{array}{cccc} 4 & -3 & 1 & -1 \\ 8 & -1 & -1 & 3 \\ 12 & 1 & -1 & -3 \\ 16 & 3 & 1 & 1 \end{array},$$

where the first column is linear in speed and the other two columns are obtained by Gram–Schmidt orthogonalization of the square and cube of the first. The corresponding columns of the design matrix, s_1 , s_2 , and s_3 , are orthogonal to the grand mean, to each other, and to the day and run effects. The parameter estimates, standard errors, and sums of squares $\hat{\gamma}_r^2 s_r^T s_r$ for these contrasts are given in Table 9.17; note that the total sum of squares equals that for speeds in Table 9.12.

The t statistics for the linear, quadratic, and cubic effects are significant at levels about 0.01, 0.07, and 0.38 respectively, suggesting that the effect of speed on efficiency may be summarized as $\hat{\mu} + \hat{\gamma}_1 s_1 + \hat{\gamma}_2 s_2$, where $\hat{\mu}$ is the estimated grand mean in this parametrization. In terms of speed, x , $s_1 = (x - 10)/2$ and $s_2 = \{(x - 10)^2 - 20\}/16$. Since $\hat{\gamma}_2 < 0$, efficiency is maximized as a function of speed when $\hat{\gamma}_1 ds_1/dx + \hat{\gamma}_2 ds_2/dx = 0$ and the estimated best speed is $10 - 4\hat{\gamma}_1/\hat{\gamma}_2 = 6.96$ mph.

Term	Estimate	Standard error	Sum of squares
Linear, s_1	−1.24	0.329	123.26
Quadratic, s_2	−1.63	0.737	42.58
Cubic, s_3	0.31	0.330	7.75
Total			173.58

Table 9.17 Field concrete mixer data: orthogonal decomposition of sum of squares for speed into linear, quadratic, and cubic effects.

A confidence interval for the true best speed may be obtained by the delta method or using an exact argument (Exercise 9.3.3), but the t statistic for $\hat{\gamma}_2$ suggests that such a confidence interval will be imprecise. The right panel of Figure 9.5 shows the fitted quadratic curve as a function of speed. ■

9.3.3 Analysis of covariance

Analysis of covariance is intended to reduce bias or increase precision when some variables cannot be controlled by design. This may arise because the importance of these variables has been recognized only after randomization, because the randomization took them partially but not fully into account, or because their values only became available after randomization.

Suppose that a model with a design matrix X from a balanced experimental setup is to be fitted, but that additional explanatory variables contained in the matrix Z have been measured that might affect the response. The design leads us to fit the model $y = X\beta + \varepsilon$, but instead we fit $y = X\beta + Z\gamma + \varepsilon$, in the hope that inclusion of Z will increase the precision of $\hat{\beta}$. However adding Z removes balance and complicates the analysis of variance. Our interest is in the treatment effects after adjusting for Z , so for analysis of variance we fit Z before treatments and their interactions.

Example 9.13 (Cat heart data) Table 9.18 shows the results from an experiment to determine the relative potencies of eight similar cardiac drugs, labelled A–H, where A is a standard. The method used was to infuse slowly a suitable dilution of the drug into an anaesthetized cat. The dose at which death occurred and the weight of the cat’s heart were recorded. The table shows $y = 100 \times \log$ dose in μgm , and, below, $z = 100 \times \log$ heart weight in gm. Four observers each made two determinations on each of eight days, with a Latin square design used to eliminate observer and time differences. Here z cannot be known at the start of the experiment, but might be expected to affect comparisons among the treatments; it is assumed that heart weight is unaffected by the treatments.

The left part of Table 9.19 gives the analysis of variance without adjustment for heart weight. The seven degrees of freedom for the sum of squares between rows have been decomposed into the main effects of observer and time, and their interaction. There are clearly large differences among observers, and between times, and a smaller but substantial interaction between these terms, but there is little evidence of day-to-day variation.

Table 9.18 Data from Latin square experiment on the potencies of eight cardiac drugs given to anaesthetized cats. The table shows $y = 100 \times \log$ dose in μgm at which death occurred, and, below, $z = 100 \times \log$ heart weight in gm.

Observer	Time	Day															
		1		2		3		4		5		6		7		8	
1	am	G	75	F	77	A	52	E	71	C	65	D	47	B	37	H	63
			91		77		102		102		84		85		73		84
1	pm	E	81	D	58	G	74	A	54	F	62	H	69	C	59	B	59
			76		90		116		87		93		79		105		71
2	am	H	94	G	86	F	104	C	66	E	94	B	72	D	82	A	58
			90		100		102		108		97		90		96		90
2	pm	B	73	A	59	E	103	F	86	G	84	C	82	H	95	D	65
			88		82		94		77		88		106		89		83
3	am	A	22	C	36	B	39	D	32	H	43	E	67	G	52	F	34
			90		81		83		94		95		101		97		66
3	pm	C	46	H	25	D	52	G	59	B	42	A	54	F	77	E	69
			90		81		91		99		90		98		106		101
4	am	F	39	E	56	H	56	B	28	D	52	G	86	A	45	C	70
			83		88		95		79		87		100		84		117
4	pm	D	87	B	82	C	72	H	92	A	58	F	92	E	99	G	89
			96		93		87		89		87		92		90		106

Table 9.19 Analysis of variance for cats data, with and without adjustment for heart weight.

Term	Without adjustment			With adjustment		
	df	Sum of squares	Mean square	df	Sum of squares	Mean square
Heart weight				1	3058	3058
Observer	3	9949	3316	3	9452	3151
Time	1	2003	2003	1	1939	1939
Observer $\times$ Time	3	2238	746	3	1890	630
Day	7	922.9	131.8	7	463	66.3
Drug	7	6098	871.1	7	5051	721.6
Residual	42	4874	116.0	41	4228	103.1

Table 9.20 Estimated differences between standard drug A and the treatments B–H, without and with adjustment for the covariate heart weight.

	B	C	D	E	F	G	H
Unadjusted	3.75 (5.4)	11.75 (5.4)	9.13 (5.4)	29.75 (5.4)	21.12 (5.4)	25.37 (5.4)	16.87 (5.4)
Adjusted	6.53 (5.2)	8.71 (5.2)	9.02 (5.1)	28.2 (5.1)	22.38 (5.1)	21.34 (5.3)	17.82 (5.1)

The estimates of the differences between the drugs and the standard, unadjusted for heart weight, are given in the upper row of Table 9.20. Their standard errors are equal because of the balance. The dose of drug B needed to cause death appears not to differ from the standard, those for C, D and H are rather larger, and those required for E, F, and G are substantially larger.

The analysis of variance with z is given in the right part of Table 9.19. Since interest centres on the drug effects, heart weight must be fitted before the term for drugs, but

	A	B	C
x	1.5, 2.2, 2.9, 4.1, 4.1	2.7, 3.8, 5.6, 6.4, 6.8	2.2, 3.5, 4.6, 5.5, 6.6
y	9.6, 11.3, 10.3, 12.5, 12.6	8.6, 7.2, 8.9, 11.6, 11.5	4.8, 5.6, 6.2, 7.5, 6.8

Table 9.21 Data from a completely randomized experiment on the comparison of diets, with initial weight x and final weight y .

otherwise it is immaterial when it is fitted; here we fit it before allowing for the experimental conditions. This results in a non-unique analysis of variance: the order of fitting is irrelevant in the left part of the table, but matters in the right part. The adjustment reduces the sums of squares for the other terms, with the reduction for days being largest. The estimate of σ^2 adjusted for heart weight, 103.1, is somewhat smaller than the unadjusted estimate, 116.0, and the precision of the comparisons between the drugs and the standard is slightly increased. In particular the adjusted estimates for B, C, and D, and for F, G, and H, are more similar — some of the variation in the unadjusted comparisons is due to heart weights. ■

Exercises 9.3

- Suppose that a 2^2 factorial experiment is to be performed using eight units in four blocks of two units each. Show that the intercept, three block effects, and the main effects and interaction between the treatments can be estimated if the treatments are allocated to blocks as follows: $(1, a)$, (b, ab) , $(1, ab)$, (a, b) . Can they all still be estimated if an observation from the last block is lost?
- Table 9.21 gives results from a completely randomized experiment in which five individuals were allocated at random to each of three diets.
 - Calculate the group averages and variances, and hence obtain the analysis of variance of the final weights, unadjusted for initial weights. Give the standard errors for differences between averages for diet A and the other two diets.
 - Use analysis of covariance to adjust for the initial weights. Give the new analysis of variance table, and adjusted standard errors for the differences in (a). Comment.
- Consider the calculation of a 95% confidence interval for the speed that gives the maximum efficiency for the field concrete mixer of Example 9.12.
 - Use the delta method to show that

$$\text{var} \left(\frac{\hat{\gamma}_1}{\hat{\gamma}_2} \right) \doteq \frac{\gamma_1^2}{\gamma_2^2} \left(\frac{\text{var}(\hat{\gamma}_1)}{\gamma_1^2} + \frac{\text{var}(\hat{\gamma}_2)}{\gamma_2^2} \right).$$

Use this to show that $\hat{\gamma}_1/\hat{\gamma}_2$ has standard error 1.595, and hence give an approximate 95% confidence interval for $10 - 4\gamma_1/\gamma_2$.

(b) If $\psi = -\gamma_1/\gamma_2$, show that the distribution of $\hat{\gamma}_2\psi + \hat{\gamma}_1$ is normal with mean zero and variance $\sigma^2(\psi^2 v_{22} + v_{11})$, where v_{11} and v_{22} are the diagonal elements of the matrix $(X^T X)^{-1}$ that correspond to γ_1 and γ_2 . Deduce that as $(\hat{\gamma}_2\psi + \hat{\gamma}_1)^2 / \{s^2(\psi^2 v_{22} + v_{11})\}$ has an $F_{1,v}$ distribution, an exact confidence region for ψ is the set of values such that $(\hat{\gamma}_2\psi + \hat{\gamma}_1)^2 / \{s^2(\psi^2 v_{22} + v_{11})\} \leq F_{1,v}(1 - \alpha)$.

A 95% confidence set for $10 + 4\psi$ based on the calculations in Example 9.12 is $(-\infty, 9.15)$, $(38.03, \infty)$. On the same graph, plot this confidence set, the average efficiencies for the different speeds and the fitted efficiency from Example 9.12 against speed. Do you find the exact confidence set surprising?

(c) Use part (b) to calculate the exact coverage of your delta method confidence interval.

9.4 Components of Variance

9.4.1 Basic ideas

Our models so far have involved just one level of random variation, with all the responses independent. Sometimes a more complex error structure is required.

The simplest example is the one-way layout with R units in each of T blocks. Suppose the blocking factors are of no intrinsic interest, and the block effects may be thought of as being sampled at random from a population, block means being a random sample from a normal distribution with mean μ and variance σ_b^2 . Conditional on the block mean, the responses for units within a block are independent normal variables with mean zero and variance σ^2 . Thus the response for the r th unit in block t is

$$y_{tr} = \mu + b_t + \varepsilon_{tr}, \quad (9.8)$$

where the b_t have zero means and variances σ_b^2 , the ε_{tr} have zero means and variances σ^2 , and the b_t and ε_{tr} are all mutually independent. Responses from different blocks are independent, but those within the same block are not, as $\text{cov}(y_{tr}, y_{ts}) = \sigma_b^2$ for $r \neq s$. Thus the covariance matrix for the responses is block diagonal. This is called a *random effects model*, as the block effects are regarded as random variables rather than fixed parameters.

The analysis of variance for the one-way layout involves the sums of squares within and between blocks

$$SS_w = \sum_{t,r} (y_{tr} - \bar{y}_{t.})^2, \quad SS_b = \sum_{t,r} (\bar{y}_{t.} - \bar{y}_{..})^2.$$

Under the random effects model, $y_{tr} - \bar{y}_{t.} = \varepsilon_{tr} - \bar{\varepsilon}_{t.}$, and as this does not depend on the presence of the random effects, SS_w has its usual $\sigma^2 \chi_{T(R-1)}^2$ distribution. Now $\bar{y}_{t.} = \mu + b_t + \bar{\varepsilon}_{t.} \sim N(\mu, \sigma_b^2 + \sigma^2/R)$, and as the $\bar{y}_{t.}$ are independent, the distribution of SS_b is $R(\sigma_b^2 + \sigma^2/R) \chi_{T-1}^2$. Furthermore,

$$\text{cov}(y_{tr} - \bar{y}_{t.}, \bar{y}_{t.} - \bar{y}_{..}) = \text{cov}(b_t + \varepsilon_{tr} - b_t - \bar{\varepsilon}_{t.}, b_t + \bar{\varepsilon}_{t.} - \bar{b}_{..} - \bar{\varepsilon}_{..}) = 0,$$

and hence the linear combinations of normal variables $y_{tr} - \bar{y}_{t.}$ and $\bar{y}_{t.} - \bar{y}_{..}$ must be independent. Thus the sums of squares SS_w and SS_b have independent chi-squared distributions with scale parameters σ^2 and $\sigma^2 + R\sigma_b^2$ respectively. Tests and confidence intervals for the ratio σ_b^2/σ^2 can be based on the $F_{T-1, T(R-1)}$ distribution of

$$\frac{\sigma^2}{\sigma^2 + R\sigma_b^2} \times \frac{SS_b/(T-1)}{SS_w/\{T(R-1)\}}. \quad (9.9)$$

An alternative derivation of the independence of SS_w and SS_b under the random effects model is to argue conditionally on the values of the b_t . Conditional on the b_t , the model is just the one-way layout described in Section 9.2.1, under which SS_w and SS_b are independent, and only the distribution of SS_b depends on the b_t . Hence SS_w and SS_b are unconditionally independent.

One aspect of interest may be statements of uncertainty for the population mean μ , which is estimated by the overall sample average, $\bar{y}_{..} = \mu + \bar{b}_{..} + \bar{\varepsilon}_{..}$. This has

Subject					
1	2	3	4	5	6
68	49	41	33	40	30
42	52	40	27	45	42
69	41	26	48	50	35
64	56	33	54	41	44
39	40	42	42	37	49
66	43	27	56	34	25
29	20	35	19	42	45

Table 9.22 Blood data: seven measurements from each of six subjects on a property related to the stickiness of their blood.

variance $\sigma_b^2/T + \sigma^2/(TR) = (\sigma^2 + R\sigma_b^2)/(TR)$, which is estimated unbiasedly by $SS_b/\{(T-1)TR\}$, independent of $\bar{y}_{..}$, and confidence intervals are based on the t_{T-1} distribution of $(\bar{y}_{..} - \mu)/[SS_b/\{(T-1)TR\}]^{1/2}$.

The assumptions of homogeneous variance across all blocks and of normality can be checked using probability plots.

Example 9.14 (Blood data) Six subjects were selected at random from a large population, and a property related to stickiness of samples of blood was measured seven times on each subject. The data are given in Table 9.22.

For these data, $SS_w = 4549.7$ and $SS_b = 1466.0$ on 36 and 5 degrees of freedom respectively. A point estimate of the variance for different measurements on the same subject is $SS_w/36 = 126.4$, and a point estimate of the variance of mean stickiness between subjects is $(SS_b/5 - SS_w/36)/7 = 23.83$. An equi-tailed 90% confidence interval for the ratio σ_b^2/σ^2 based on (9.9) is $(-0.01, 1.34)$; this overlaps the negative half-axis and would not usually be appropriate. ■

Nested variation

The previous example had two levels of nested variation, for subjects and for measurements. In practice data with several levels of variation arise. Consider for example comparison of the success of a surgical procedure, measured on a continuous scale. Data are available on patients, P of whom are treated by each surgeon and with S surgeons working at H hospitals. We suppose that surgeons at different hospitals are independent, and likewise for the patients, so patients are nested within surgeons within hospitals — there is no relation between the first patient of surgeon 1 at hospital 1 and the first patient of surgeon 2 at hospital 1, nor between surgeon 1 at hospital 1 and surgeon 1 at hospital 2. Put another way, labels for patients can be permuted independently within each surgeon without changing the data structure, and likewise for surgeons within each hospital. A simple model for the outcome y_{hsp} for the p th patient of the s th surgeon at the h th hospital is

$$y_{hsp} = \mu + b_h + e_{hs} + \varepsilon_{hsp}, \quad h = 1, \dots, H, s = 1, \dots, S, p = 1, \dots, P, \quad (9.10)$$

Term	df	Sum of squares	E(Mean square) when terms below random		
			ε	ε, e	ε, e, b
Between hospitals	$H - 1$	$\sum (\bar{y}_{h..} - \bar{y}_{...})^2$	$PS\delta_b^2 + P\delta_e^2 + \sigma^2$	$PS\delta_b^2 + P\sigma_e^2 + \sigma^2$	$PS\sigma_b^2 + P\sigma_e^2 + \sigma^2$
Between surgeons within hospitals	$H(S - 1)$	$\sum (\bar{y}_{hs.} - \bar{y}_{h..})^2$	$P\delta_e^2 + \sigma^2$	$P\sigma_e^2 + \sigma^2$	$P\sigma_e^2 + \sigma^2$
Between patients within surgeons	$HS(P - 1)$	$\sum (y_{hsp} - \bar{y}_{hs.})^2$	σ^2	σ^2	σ^2

Table 9.23 Analysis of variance table for nested model. Each sum of squares is summed over h , s and p . Mean squares are formed by dividing sums of squares by their degrees of freedom. δ_b^2 and δ_e^2 are non-centrality parameters measuring differences among the b_h and e_{hs} when they are treated as fixed.

where μ is the mean success level in a population of hospitals, from which the h th hospital departs by b_h , the e_{hs} represent surgeon effects, and the ε_{hsp} are independent normal variables with means zero and variance σ^2 corresponding to the p th patient treated by the s th surgeon at hospital h . If random, we suppose the b_h and e_{hs} to be independent normal variables with means zero and variances σ_b^2 and σ_e^2 , but the decision whether they should be treated as random or as fixed depends on the context. A potential patient able to choose his surgeon would treat b_h and e_{hs} as fixed, and hope to choose h and s to optimize his prospects. If on the other hand he could choose his hospital but not his surgeon, he might treat the e_{hs} as random — in effect he will be operated upon by a randomly selected surgeon — but try and choose among hospitals, treated as fixed. A health service official hoping to estimate the national success rate for the procedure from a sample of such data would treat the b_h and the e_{hs} as random. The quantities of interest in the three cases are $\mu + b_h + e_{hs}$, $\mu + b_h$, and μ , estimated by $\bar{y}_{hs.}$, $\bar{y}_{h..}$, and $\bar{y}_{...}$, whose variances are σ^2/P , $\sigma_e^2/S + \sigma^2/(SP)$, and $\sigma_b^2/H + \sigma_e^2/(HS) + \sigma^2/(HSP)$. In each case the analysis of variance is given by Table 9.23 and depends on

$$\begin{aligned}\bar{y}_{h..} - \bar{y}_{...} &= b_h - \bar{b}_{..} + \bar{e}_{h.} - \bar{e}_{..} + \bar{\varepsilon}_{h..} - \bar{\varepsilon}_{...}, \\ \bar{y}_{hs.} - \bar{y}_{h..} &= e_{hs} - \bar{e}_{h.} + \bar{\varepsilon}_{hs.} - \bar{\varepsilon}_{h..}, \\ y_{hsp} - \bar{y}_{hs.} &= \varepsilon_{hsp} - \bar{\varepsilon}_{hs.}.\end{aligned}$$

If all the quantities contributing to it are regarded as random, then each sum of squares has a chi-squared distribution. For example, the sum of squares between surgeons within hospitals is

$$SS_S = \sum_{h,s,p} (\bar{y}_{hs.} - \bar{y}_{h..})^2 = P \sum_{h,s} (e_{hs} - \bar{e}_{h.} + \bar{\varepsilon}_{hs.} - \bar{\varepsilon}_{h..})^2,$$

and if e_{hs} and ε_{hsp} are random, then $e_{hs} + \bar{\varepsilon}_{hs.}$ is normal with mean zero and variance $\sigma_e^2 + \sigma^2/P$. Hence

$$\begin{aligned}P \sum_{h,s} (e_{hs} - \bar{e}_{h.} + \bar{\varepsilon}_{hs.} - \bar{\varepsilon}_{h..})^2 &\stackrel{D}{=} P(\sigma_e^2 + \sigma^2/P)(W_1 + \cdots + W_H) \\ &\sim (P\sigma_e^2 + \sigma^2)\chi_{H(S-1)}^2,\end{aligned}$$

where the W_h are a random sample from the χ^2_{S-1} distribution. If the e_{hs} are fixed, then $e_{hs} + \bar{e}_{hs}$ is normal with mean e_{hs} and variance σ^2/P and hence SS_S has a non-central chi-squared distribution with $H(S-1)$ degrees of freedom and non-centrality parameter $H(S-1)\delta_e^2 = P \sum_{h,s} (e_{hs} - \bar{e}_h)^2$ (Problem 2.12). Such calculations give the entries in Table 9.23, in which $(H-1)\delta_b^2 = \sum_h (b_h - \bar{b})^2$. Note that $E(\delta_e^2) = \sigma_e^2$ and $E(\delta_b^2) = \sigma_b^2$.

Under the model with b_h and e_{hs} fixed, ratios of mean squares can be used to test for differences among surgeons and hospitals, for example comparing the ratio of mean squares for the last two lines of Table 9.23 with the $F_{H(S-1), HS(P-1)}$ distribution.

The assumptions underlying this model would need careful scrutiny in applications: from what populations are patients, surgeons, and hospitals drawn, and in what sense can they be treated as random samples?

Nesting is fundamentally different from the type of classification described earlier. Consider a two-way layout in which factors A and B with T and R levels respectively are applied to TR units. Then if the levels of B among $y_{11}, \dots, y_{1R}$ were permuted, the same permutation would have to be applied to those of $y_{t1}, \dots, y_{tR}$ for each t , because the second subscript corresponds to the same treatment for y_{2r} and y_{1r} , for example. The two classifications are then said to be *crossed*. In the random effects model described at the start of this section, however, the labelling is essentially arbitrary, $y_{t1}, \dots, y_{tR}$ being simply replicate observations; here permutation of any or all of these groups of observations should not affect analysis. Compare Examples 9.3 and 9.14.

It is crucial that crossed and nested effects be distinguished. Typically the levels of crossed effects are of intrinsic interest and are represented by fixed parameters, while parameters associated with nested suffixes are treated as random. Different levels of nesting then correspond to different variance components. However it may be hard to write down the model appropriate to a complex design.

Split-unit experiments

Some experiments are performed with certain treatments applied to entire units and others to sub-units. As such designs originally arose in agriculture, with units being for instance plots of land sown with plant varieties, sub-plots of which were treated with different fertilisers, they are often called split-plot experiments. They also arise in industrial applications, where certain aspects of a manufacturing process may be more easily varied than others, and in medical settings where units are often patients, each with measurements taken in succession over a period, giving a series of correlated responses. Such designs are useful if it is already known that whole-unit treatments differ substantially and interest centres on sub-unit treatments and their interactions, or if physical constraints impose them; they can also arise by accident. The key idea is that there is variation within units (that is, between sub-units) as well as between units. Analysis of variance is effectively performed at two levels, as discussed below.

Suppose there are B blocks of W units, to each of which a whole-unit treatment is applied according to a randomized block design, for example. Units themselves are

split into S sub-units, with a sub-unit treatment randomized to each. The corresponding linear model is

$$y_{bws} = \mu + \beta_b + \gamma_w + u_{bw} + \zeta_s + \tau_{ws} + \varepsilon_{bws}, \quad b = 1, \dots, B, w = 1, \dots, W, \\ s = 1, \dots, S,$$

where the whole-unit effects are the overall mean μ , the block and whole-unit treatment parameters β_b and γ_w , and the whole-unit errors u_{bw} , taken to be independent normal variables with mean zero and variance σ_u^2 . The sub-unit treatment effects are ζ_s , the interactions between sub- and whole-unit treatments τ_{ws} and the sub-unit errors ε_{bws} , taken to be normal with mean zero and variance σ^2 independent of each other and of the u_{bw} . Terms ξ_{bs} are not included because interaction between blocks and sub-units makes no sense.

Under this model different treatments are analyzed at different levels. Whole-unit averages have variance $\sigma_u^2 + \sigma^2/S$ estimated by the residual mean square from a randomized block analysis of these averages, with $(B-1)(W-1)$ degrees of freedom. Whole-unit treatments are compared using contrasts of these averages such as

$$\bar{y}_{..2} - \bar{y}_{..1} = \gamma_2 - \gamma_1 + \bar{u}_{..2} - \bar{u}_{..1} + \bar{\varepsilon}_{..2} - \bar{\varepsilon}_{..1},$$

whose variance is $2\sigma_u^2/B + 2\sigma^2/(BS)$. Comparisons of sub-unit treatments and their interactions with whole-unit treatments use the $BW(S-1)$ remaining degrees of freedom and involve quantities such as

$$\bar{y}_{..2} - \bar{y}_{..1} = \zeta_2 - \zeta_1 + \bar{\tau}_{..2} - \bar{\tau}_{..1} + \bar{\varepsilon}_{..2} - \bar{\varepsilon}_{..1}, \\ (\bar{y}_{.22} - \bar{y}_{.21}) - (\bar{y}_{.12} - \bar{y}_{.11}) = \tau_{22} - \tau_{21} - \tau_{12} + \tau_{11} + \bar{\varepsilon}_{.22} - \bar{\varepsilon}_{.21} - \bar{\varepsilon}_{.12} + \bar{\varepsilon}_{.11},$$

with variances respectively $2\sigma^2/(BW)$ and $4\sigma^2/B$. As there are $S-1$ degrees of freedom for sub-unit treatments and $(S-1)(W-1)$ degrees of freedom for their interactions with whole-unit treatments,

$$BW(S-1) - (S-1) - (S-1)(W-1) = W(B-1)(S-1)$$

degrees of freedom remain for estimation of σ^2 . If the variability between whole units is larger than that within them, that is, $\sigma^2 < \sigma_u^2$, then comparisons among sub-unit treatments and their interactions with whole-unit treatments will be more precise than among whole-unit treatments themselves.

Example 9.15 (Cake data) Table 9.24 gives data from an experiment in which six different temperatures for cooking three recipes for chocolate cake were compared. Each time a mix was made using one of the recipes, enough batter was prepared for six cakes, which were then randomly allocated to be cooked at temperatures 175, 185, ..., 225°C. Thus mixes correspond to blocks, recipes are the whole-unit treatments and baking temperatures the sub-unit treatments. We suppose that the 15 mixes of each recipe were made in order 1, ..., 15, so that mix is a surrogate for time.

The response is the breaking angle, found by fixing one half of a slab of cake, then pivoting the other half about the middle until breakage occurs. Let y_{rmt} denote the response for the r th recipe, m th mixture and t th temperature, where $r = 1, \dots, 3$,

We use the Roman letter u to indicate that the whole-unit effects are regarded as random variables rather than as parameters.

Recipe	Temp °C	Mix														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	175	42	47	32	26	28	24	26	24	24	24	33	28	29	24	26
	185	46	29	32	32	30	22	23	33	27	33	39	31	28	40	28
	195	47	35	37	35	31	22	25	23	28	27	33	27	31	29	32
	205	39	47	43	24	37	29	27	32	33	31	28	39	29	40	25
	215	53	57	45	39	41	35	33	31	34	30	33	35	37	40	37
	225	42	45	45	26	47	26	35	34	23	33	30	43	33	31	33
2	175	39	35	34	25	31	24	22	26	27	21	20	23	32	23	21
	185	46	46	30	26	30	29	25	23	26	24	27	28	35	25	21
	195	51	47	42	28	29	29	26	24	32	24	33	31	30	22	28
	205	49	39	35	46	35	29	26	31	28	27	31	34	27	19	26
	215	55	52	42	37	40	24	29	27	32	37	28	31	35	21	27
	225	42	61	35	37	36	35	36	37	33	30	33	29	30	35	20
3	175	46	43	33	38	21	24	20	24	24	26	28	24	28	19	21
	185	44	43	24	41	25	33	21	23	18	28	25	30	29	22	28
	195	45	43	40	38	31	30	31	21	21	27	26	28	43	27	25
	205	46	46	37	30	35	30	24	24	26	27	25	35	28	25	25
	215	48	47	41	36	33	37	30	21	28	35	38	33	33	25	31
	225	63	58	38	35	23	35	33	35	28	35	28	28	37	35	25

Table 9.24 Data on breaking angles (°) of chocolate cakes (Cochran and Cox, 1959, p. 300).

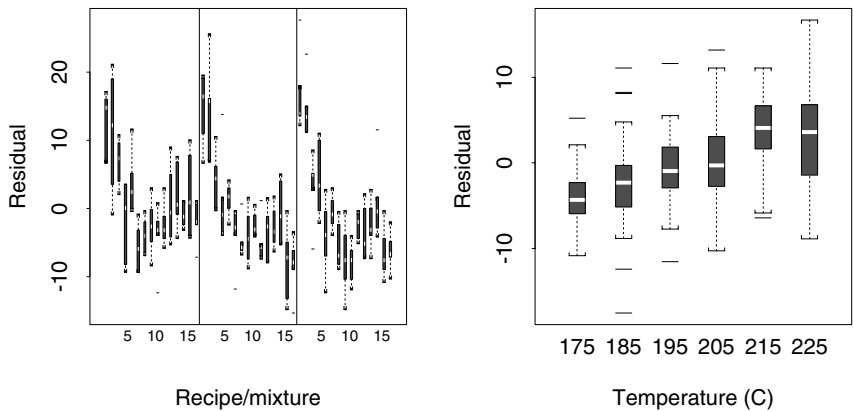


Figure 9.8 Cake data. Left: variation of $y_{rmt} - \bar{y}_{rm}$, across mixes for the three recipes. The vertical lines demarcate results for the three recipes. Right: dependence of $y_{rmt} - \bar{y}_{.t}$ on temperature.

$m = 1, \dots, 15$ and $t = 1, \dots, 6$. The model we consider is

$$y_{rmt} = \mu + \beta_r + \gamma_m + u_{rm} + \zeta_t + \tau_{rt} + \varepsilon_{rmt}, \quad (9.11)$$

where the $u_{rm} \stackrel{\text{iid}}{\sim} N(0, \sigma_u^2)$ represent the whole-unit errors corresponding to the mixes made for each recipe, and the $\varepsilon_{rmt} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ denote the sub-unit errors. We treat the β_r and γ_m as parameters and the u_{rm} as random variables because if the experiment was repeated, the recipes would be unchanged and the time ordering would still arise, but the mixes would be different.

The left panel of Figure 9.8 shows how $y_{rmt} - \bar{y}_{rm}$ varies across mixes for the three recipes. There is evidently a systematic effect of mix, with responses for the first few

Table 9.25 Analysis of variance on a split-unit basis for cakes data. The F statistics for the upper part are computed using the residual sum of squares (a) for contrasts among whole units. Those in the lower part are computed using (b), the residual sum of squares for contrasts among split units. There are large differences among mixes and temperatures, but not among recipes. The temperature effect is essentially linear.

Source of variation	df	Sum of squares	Mean square	F
Mixes	14	8.159	0.583	12.15
Recipes	2	0.186	0.093	1.93
Residual (a)	28	1.343	0.048	
Temperatures	5	2.051	0.410	21.32
Linear	1	1.925	1.925	100.08
Quadratic	1	0.021	0.021	1.08
Cubic, quartic, quintic	3	0.105	0.035	1.82
Recipes $\times$ Temperatures	10	0.176	0.018	0.91
Residual (b)	210	4.040	0.019	

Table 9.26 Average log breaking angle (degrees) of cakes by recipe and temperature.

Recipe	Temperature ($^{\circ}\text{C}$)						Average
	175	185	195	205	215	225	
1	3.350	3.433	3.409	3.493	3.638	3.535	3.476
2	3.270	3.355	3.428	3.443	3.505	3.537	3.423
3	3.293	3.331	3.428	3.405	3.516	3.538	3.419
Average	3.304	3.373	3.422	3.447	3.553	3.537	3.439

mixes for each recipe substantially greater than for later ones, but any recipe differences seem small. The right panel shows roughly linear dependence on temperature of the differences $y_{rmt} - \bar{y}_{..t}$, from which whole-unit variation has been eliminated, but there is a perceptible increase in variance with mean. This suggests use of log-transformed responses, which is confirmed by a Box–Cox analysis (Example 8.23).

Table 9.25 shows the analysis of variance for the model fitted to the log responses. There are $3 \times 15 = 45$ whole units, from which a grand mean and 44 contrasts may be computed. The component of variance for these 44 degrees of freedom is shown in the upper part of the table, split into 14 degrees of freedom among mixes, 2 among recipes and 28 residuals. The mean square at (a) estimates $\sigma^2 + 6\sigma_u^2$ and is the appropriate basis for comparison of recipes and mixes. There are large differences among mixes but not among recipes.

The lower part of the table shows the $3 \times 15 \times (6 - 1) = 225$ degrees of freedom for contrasts within whole units, of which there are 5 among temperatures and $(3 - 1) \times (6 - 1)$ for the recipe $\times$ temperature interaction. The mean square for residual (b) estimates σ^2 , and comparison with (a) gives estimate $(0.048 - 0.019)/6 = 0.0035$ of σ_u^2 , rather small variation among mixes. A split of the overall temperature effect into linear, quadratic and remaining effects confirms the linearity of the effect of temperature on the response.

Table 9.26 shows average log breaking angles by recipe and temperature. Each average is based on 15 raw observations and has variance $\sigma_u^2 + \sigma^2/15$, but while

differences between rows involve the u 's, those between columns do not. Differences between two recipe averages and between two temperature averages have variances $2(\sigma_u^2 + \sigma^2/6)/15$ and $2(\sigma^2/15)/3$, estimated by $2 \times 0.048/90 = 0.033^3$ and $2 \times 0.019/45 = 0.029^2$. The difference between two temperature averages for one recipe, $\bar{y}_{r \cdot t_1} - \bar{y}_{r \cdot t_2}$, does not depend on the u_{rm} , so its variance is $2\sigma^2/15$, while the difference of two recipe averages for a given temperature, $\bar{y}_{r_1 \cdot t} - \bar{y}_{r_2 \cdot t}$, involves both u 's and ε 's and has variance $2(\sigma_u^2 + \sigma^2)/15$; these variances are estimated respectively by $2 \times 0.019/15 = 0.050^2$ and $2 \times \{5 \times 0.019 + 0.048\}/90 = 0.056^2$.

The best summary of the results here is Table 9.26, supplemented by the standard errors for comparisons among the averages. ■

9.4.2 Linear mixed models

In many situations the comparison of treatments is complicated by correlations among the responses. In medical settings, for example, a common design involves repeated measures on the same individual, leading to *repeated measures* or *longitudinal* data. Related designs arise in many types of investigation. Although the notion of levels of variation underlying the classical split-plot experiment remains very useful, such data are rarely neatly balanced and their analysis and interpretation is less straightforward. In this section we briefly put such experiments in a more general context.

When confronted with a complex experiment, it is helpful to ask if it is reasonable to assume that the levels of certain factors have been selected from a population. If so, we ask if interest resides purely in the population, or also in the realized values of random variables sampled from it. When this latter is the case, then we must estimate not only properties of the population but also the underlying variables. In dairy herd breeding experiments, for example, bulls and cows are mated and the milk yield of their daughters is treated as the response. As any repetition of the experiment would involve different animals, they are regarded as randomly sampled from a population. It is useful to estimate effects for individual animals, however, in order to retain for future breeding those bulls whose daughters give the best yield. Thus although a random effects model is appropriate, estimates of the random effects are required. Similar considerations arise in many other contexts, and we now discuss inference for random effects.

We consider normal linear models of form

$$y = X\beta + Zb + \varepsilon, \quad (9.12)$$

where in addition to the usual setup the $n \times q$ matrix Z indicates how the response vector y depends on the $q \times 1$ vector of unobserved random variables b . This is called a *mixed model* because the response depends on random variables b as well as on fixed parameters β . If b is normal with mean zero and covariance matrix Ω_b , then we may write

$$y | b \sim N_n(X\beta + Zb, \Omega) \quad \text{and} \quad b \sim N_q(0, \Omega_b). \quad (9.13)$$

Thus the marginal density of y is normal with mean $X\beta$ and variance matrix $Z\Omega_b Z^T + \Omega$, which does not depend on β . In most cases $\Omega = \sigma^2 I_n$, where $\sigma^2 = \text{var}(\varepsilon_j)$, and later it will be useful to write $Z\Omega_b Z^T + \Omega = \sigma^2 \Upsilon^{-1}$, say. We use ψ to denote the vector of distinct variance ratios appearing in Υ^{-1} .

Example 9.16 (Longitudinal data) A short longitudinal study has one individual allocated to the treatment and two to the control, with observations

$$y_{1j} = \beta_0 + b_1 + \varepsilon_{1j}, \quad y_{21} = \beta_0 + b_2 + \varepsilon_{21}, \quad y_{3j} = \beta_0 + \beta_1 + b_3 + \varepsilon_{3j}, \quad j = 1, 2.$$

Thus there are two measurements on the first and third individuals, and just one on the second. The b_j represent variation among individuals and the ε_{ij} variation between measures on the same individuals. If the b 's and ε 's are all mutually independent with variances σ_b^2 and σ^2 , then

$$\begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{31} \\ y_{32} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} + \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{21} \\ \varepsilon_{31} \\ \varepsilon_{32} \end{pmatrix},$$

and this fits into formulation (9.12) with $\Omega_b = \sigma_b^2 I_3$ and $\Omega = \sigma^2 I_5$. Here ψ comprises the scalar σ_b^2/σ^2 , and hence the variance matrix

$$\Omega + Z\Omega_b Z^T = \begin{pmatrix} \sigma_b^2 + \sigma^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 + \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & \sigma_b^2 + \sigma^2 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2 + \sigma^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 + \sigma^2 \end{pmatrix}$$

may be written as

$$\sigma^2 \Upsilon^{-1} = \sigma^2 \begin{pmatrix} 1 + \psi & \psi & 0 & 0 & 0 \\ \psi & 1 + \psi & 0 & 0 & 0 \\ 0 & 0 & 1 + \psi & 0 & 0 \\ 0 & 0 & 0 & 1 + \psi & \psi \\ 0 & 0 & 0 & \psi & 1 + \psi \end{pmatrix},$$

of block diagonal form. ■

In principle likelihood inference for the parameters of this model may be based on the marginal normal density of y , which gives log likelihood

$$\ell(\beta, \sigma^2, \psi) \equiv -\frac{1}{2\sigma^2}(y - X\beta)^T \Upsilon (y - X\beta) - \frac{n}{2} \log \sigma^2 + \frac{1}{2} \log |\Upsilon|,$$

where Υ depends on ψ . For known ψ the maximum likelihood estimators of β and σ^2 are

$$\hat{\beta}_\psi = (X^T \Upsilon X)^{-1} X^T \Upsilon y, \quad \hat{\sigma}_\psi^2 = n^{-1} (y - X\hat{\beta})^T \Upsilon (y - X\hat{\beta}),$$

so the profile log likelihood for ψ is $\ell_p(\psi) \equiv -\frac{1}{2}n \log \hat{\sigma}_\psi^2 + \frac{1}{2} \log |\Upsilon|$. We maximize this to estimate ψ , and then obtain maximum likelihood estimates $\hat{\beta}_{\hat{\psi}}$ and $\hat{\sigma}_{\hat{\psi}}^2$. Thus inference boils down to maximization of $\ell_p(\psi)$.

Unfortunately life is not so simple. One difficulty is that the maximum likelihood variance estimators can have large downward bias because no adjustment is made for the degrees of freedom lost in estimating the $p \times 1$ vector β . In such models p can be large, and then it is important to replace the divisor n in $\hat{\sigma}^2$ by the true degrees of freedom $n - p$. Adjustment both for this and for estimation of the elements of ψ can be performed by maximizing the modified log likelihood

$$\ell(\beta, \sigma^2, \psi) + \frac{p}{2} \log \sigma^2 - \frac{1}{2} \log |X^T \Upsilon X|.$$

This procedure, known as REML or restricted maximum likelihood estimation, is justified in Section 12.2. It turns out to be equivalent to use of a marginal likelihood, that is, a likelihood formed from a cunningly chosen marginal density rather than the full density of the data.

A second difficulty is that the domain for ψ is $[0, \infty)^{\dim \psi}$. If the maximum occurs on the boundary of this set, then standard likelihood theory does not apply to confidence intervals and so forth. Care must anyway be used unless the maximum lies well away from the boundary. If so, standard errors for $\hat{\beta}$ are found from $\sigma^2(X^T \Upsilon X)^{-1}$ with parameters replaced by estimates.

$\dim \psi$ is the dimension of ψ .

A third difficulty is computational: in realistic problems the matrices involved in such models can be large enough that even specially designed optimization routines converge only slowly.

Prediction of random effects

Once estimates of β , σ^2 , and ψ have been obtained, the question arises how to perform inference for the random variables b . We prefer to reserve the term *estimation* for unknown parameters and to speak of *prediction* of unobserved random variables. In normal models it is natural to choose the predictor $\tilde{b} = \tilde{b}(y)$ to be the function of y that minimizes the mean squared prediction error

$$E[\{\tilde{b}(y) - b\}^T \{\tilde{b}(y) - b\}],$$

where the expectation is over both b and y . It is straightforward to show that this is achieved by taking $\tilde{b}(y) = E(b | y)$, the conditional mean of b given y . As b and y have a joint normal distribution, we obtain (Exercise 9.4.5)

$$E(b | y) = (Z^T \Omega^{-1} Z + \Omega_b^{-1})^{-1} Z^T \Omega^{-1} (y - X\beta), \quad (9.14)$$

$$\text{var}(b | y) = (Z^T \Omega^{-1} Z + \Omega_b^{-1})^{-1}. \quad (9.15)$$

Replacement of the unknown parameters by estimates results in the predictions $\tilde{b}$ and their estimated variance. It turns out that the $\tilde{b}$ are best linear unbiased predictors (Problem 9.6). If Ω_b^{-1} was absent, then (9.14) would be the weighted least squares estimator from regressing $(y - X\beta)$ on the columns of Z with weight matrix Ω^{-1} .

They are often called BLUPs.

The presence of Ω_b^{-1} means that $\tilde{b}$ is shrunk towards zero compared to the weighted least squares estimator, and for this reason $\tilde{b}$ is known as a *shrinkage* estimator.

The residuals too are modified due to shrinkage. As

$$\begin{aligned} y - X\hat{\beta} &= Z\tilde{b} + y - X\hat{\beta} - Z\tilde{b} \\ &= Z\tilde{b} + \{I_n - Z(Z^T\hat{\Omega}^{-1}Z + \hat{\Omega}_b^{-1})^{-1}Z^T\hat{\Omega}^{-1}\}(y - X\hat{\beta}), \end{aligned}$$

the residuals $y - X\hat{\beta}$ split into two parts, the first $Z\tilde{b}$ being attributable to the predicted random effects, and the second being the usual residual $y - X\hat{\beta}$ shrunk towards zero; this estimates ε .

Example 9.17 (One-way layout) Consider the unbalanced one-way layout model

$$y_{ij} = \mu + b_i + \varepsilon_{ij}, \quad j = 1, \dots, n_i, \quad i = 1, \dots, q,$$

in which the group effects $b_i \stackrel{\text{iid}}{\sim} N(0, \sigma_b^2)$ independently of the individual errors $\varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$. This generalizes (9.8). In terms of (9.12),

$$\Omega = \sigma^2 I_n, \quad \Omega_b = \sigma_b^2 I_q, \quad X = 1_n, \quad Z = \begin{pmatrix} 1_{n_1} & 0 & \cdots & 0 \\ 0 & 1_{n_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1_{n_q} \end{pmatrix},$$

where $n = n_1 + \cdots + n_q$. Substitution into (9.14) and (9.15) reveals that the i th element of $\tilde{b}$ and its estimated variance are

$$\tilde{b}_i = \frac{\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}}{1 + \hat{\sigma}^2 / (n_i \hat{\sigma}_b^2)}, \quad \frac{1}{1/\hat{\sigma}_b^2 + n_i/\hat{\sigma}^2},$$

so the fixed-effects estimator $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ is shrunk towards zero by an amount that depends on the estimated variance ratio. The shrinkage will be considerable if $\hat{\sigma}^2/n_i \gg \hat{\sigma}_b^2$, corresponding to large variation in the group averages owing to individual variances compared to the variation between groups, as in Example 9.14. The data are then almost a simple random sample of size n , so strong shrinkage is not surprising.

The variance formula is also instructive, as $\text{var}(\tilde{b}_i | y) \rightarrow 0$ when $\sigma_b^2 \rightarrow 0$, $\sigma^2 \rightarrow 0$, or $n_i \rightarrow \infty$. In the first case, there is no variation between groups, and hence $b_i = 0$ with probability one. In the second two cases, the value of b_i is known exactly, because variation around it is negligible. The practical implication is that consistent inference for b_i is impossible when σ_b^2 and σ^2 take positive values: even if $q \rightarrow \infty$, the amount of information on any given b_i does not accumulate unless $n_i \rightarrow \infty$, and this is rarely the case. This applies to estimation of random effects more generally. ■

Example 9.18 (Rat growth data) Table 9.27 gives the weights of $n = 30$ young rats measured for five weeks. The left panel of Figure 9.9 shows that although the weight of each rat grows roughly linearly, neither slope nor intercept appears to be common to all the animals. This is confirmed by the analysis of variance from fitting standard linear models with common intercept and slope, different intercepts, and both intercepts and slopes different: the F tests are all highly significant.

	Week						Week				
	1	2	3	4	5		1	2	3	4	5
1	151	199	246	283	320	16	160	207	248	288	324
2	145	199	249	293	354	17	142	187	234	280	316
3	147	214	263	312	328	18	156	203	243	283	317
4	155	200	237	272	297	19	157	212	259	307	336
5	135	188	230	280	323	20	152	203	246	286	321
6	159	210	252	298	331	21	154	205	253	298	334
7	141	189	231	275	305	22	139	190	225	267	302
8	159	201	248	297	338	23	146	191	229	272	302
9	177	236	285	340	376	24	157	211	250	285	323
10	134	182	220	260	296	25	132	185	237	286	331
11	160	208	261	313	352	26	160	207	257	303	345
12	143	188	220	273	314	27	169	216	261	295	333
13	154	200	244	289	325	28	157	205	248	289	316
14	171	221	270	326	358	29	137	180	219	258	291
15	163	216	242	281	312	30	153	200	244	286	324

Table 9.27 Weights (units unknown) of 30 young rats over a five-week period (Gelfand *et al.*, 1990).

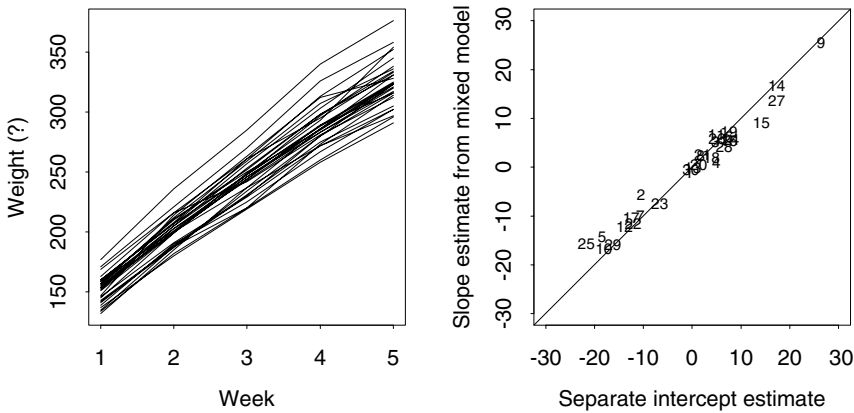


Figure 9.9 Rat growth data. Left: weekly weights of 30 young rats. Right: shrinkage of individual slope estimates towards overall slope estimate; the solid line has unit slope, and the estimates from the mixed model lie slightly closer to zero than the individual estimates.

We treat the rats as a sample from a population of similar creatures, with different initial weights and growing at different rates. To model this we express the data from the j th rat as

$$y_{jt} = \beta_0 + b_{j0} + (\beta_1 + b_{j1})x_{jt} + \varepsilon_{jt}, \quad t = 1, \dots, 5,$$

where the random variables (b_{j0}, b_{j1}) have a joint normal distribution with mean vector zero and unknown variance matrix. In matrix terms we have

$$\begin{pmatrix} y_{j1} \\ \vdots \\ y_{j5} \end{pmatrix} = \begin{pmatrix} 1 & x_{j1} \\ \vdots & \vdots \\ 1 & x_{j5} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} 1 & x_{j1} \\ \vdots & \vdots \\ 1 & x_{j5} \end{pmatrix} \begin{pmatrix} b_{j0} \\ b_{j1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{j1} \\ \vdots \\ \varepsilon_{j5} \end{pmatrix}, \quad j = 1, \dots, n,$$

and the overall model is obtained by stacking these expressions. Below we take $(x_{j1}, \dots, x_{j5}) = (0, \dots, 4)$, so that the intercept β_0 corresponds to the weight in week 1. There are just $p = 2$ population parameters β_0 and β_1 , but $q = 60$ because

Table 9.28 Results from fit of mixed model to rat growth data, using REML. Values in parentheses are for maximum likelihood fit. In each case $\hat{\sigma}^2 = 5.82^2$.

Parameter	Fixed		Random	
	Estimate	Standard error	Variance	Correlation
Intercept	156.05	2.16 (2.13)	10.93 ² (10.71 ²)	
Slope	43.27	0.73 (0.72)	3.53 ² (3.46 ²)	0.18 (0.19)

there are two random variables per rat. We assume that the within-rat errors ε_{jt} are independent normal variables with variances σ^2 , independent of the b 's.

Table 9.28 gives estimates from REML and maximum likelihood fits of this model. As expected, the maximum likelihood estimates for variances are smaller than the REML estimates, but here p is small and the difference is minimal. The estimated mean weight in week 1 is 156, but the variability from rat to rat has estimated standard deviation of about 11 about this. The slopes show similarly large variation. Correlation between the slope and intercept variables is small, however. The measurement error variance $\hat{\sigma}^2 = 5.82^2$ is smaller than is the inter-rat variation in intercepts, but exceeds that for slopes.

The right panel of Figure 9.9 shows how the slope estimates from fitting separate models to each rat are shrunk towards the overall value. The amount of shrinkage is small, owing to the relatively large variation among the rats relative to $\hat{\sigma}^2$, and as it depends on the intercepts it is not uniform.

Probability plots show that the residuals and random effects $\tilde{b}$ are reasonably close to normal, but a plot of residuals against week suggests adding quadratic terms $(\beta_2 + b_{j2})x_{jt}^2$. Their inclusion reduces AIC for REML from 1096.58 to 1013.36, a large improvement, but the resulting model involves predicting 90 b 's from 150 observations, leaving only about two observations per rat for model checking. Fortunately cubic terms do not seem to be necessary as well. ■

Sometimes it is helpful to separate b into sub-vectors corresponding to different levels of variation. For example, educational studies may involve classes of students in different schools belonging to different educational authorities, so that comparisons of outcomes must take into account different levels of random effects as well as fixed effects corresponding to types of school, socio-economic background of students, and so forth. For data with L levels of variation it may be useful to write this as a multi-level model

$$y = X\beta + Z_L b_L + \cdots + Z_0 b_0, \quad (9.16)$$

where the $q_l \times 1$ vectors b_l are all mutually independent with means zero and variance matrices Ω_l . Then the marginal mean of y is $X\beta$, while its variance is $\sum_{l=0}^L Z_l \Omega_l Z_l^T$. For consistency we set $Z_0 = I_n$ and let b_0 contain the errors, so $b_0 = \varepsilon$ and $\Omega_0 = \sigma^2 I_n$. The examples above have $L = 1$, so in addition to measurement error there is just one other level of variation, corresponding to individuals. More generally $L > 1$, and $\text{var}(y)$ is formed by adding block diagonal matrices. References to fuller discussions can be found in Section 9.5.

Doctor	Pipette and counting chamber									
	1	2	3	4	5	6	7	8	9	10
A	427	372	418	440	349	484	430	416	449	464
B	434	420	385	472	415	420	415	396	439	424
C	480	421	473	496	474	411	472	423	502	488
D	451	369	500	464	444	410	422	396	459	471
E	462	453	450	520	489	409	508	347	440	391

Table 9.29 The numbers of red blood cells counted by five doctors using ten sets of apparatus.

Exercises 9.4

- Consider (9.8).
 - Show that a confidence interval for the mean of the t th group, $\alpha_t = \mu + b_t$, may be based on $T = (\bar{y}_t - \alpha_t)/[SS_w/\{TR(R-1)\}]^{1/2}$, and give its distribution.
 - Suppose we take a single observation on a randomly selected block. Show that its variance is $\sigma^2 + \sigma_b^2$, and that this is estimated unbiasedly by $\{TSS_b + (T-1)SS_w\}/\{RT(T-1)\}$.
 - Suppose a fixed number $n = RT$ of units is available, and that it is required to minimize the variance of the population mean estimator, $\bar{y}_{..}$. Show that we should take T as large as possible, that is, $R = 2$.
 - Suppose there is a cost c_0 for measuring the response on each unit, and a cost c_1 for each group. Show that the total cost is $RTc_0 + Tc_1$, and find R to minimize $\text{var}(\bar{y}_{..})$ subject to a fixed total cost.
- Discuss how to check the assumptions of the components of variance model (9.8) using (i) normal probability plots of the $y_{tr} - \bar{y}_t$, and of the $\bar{y}_t$, and (ii) chi-squared probability plots of the group sums of squares $\sum_r (y_{tr} - \bar{y}_t)^2$. In each case give the expected slope and intercept of the plot. Show that if R is small a normal scores plot of the $(y_{tr} - \bar{y}_t)/\{(R-1)/R\}^{1/2}$ is preferable to one based on the $y_{tr} - \bar{y}_t$. Discuss whether (9.8) is appropriate for the data of Example 9.14.
- Table 9.29 gives the numbers of red blood cells counted by five doctors using ten sets of apparatus. Suppose that both doctors and sets of apparatus are thought of as randomly selected from suitable populations, and that the response for the r th doctor and c th set of apparatus is

$$y_{rc} = \mu + d_r + a_c + \varepsilon_{rc},$$

where d_r , a_c , and ε_{rc} are independent normal variables with zero means and variances σ_d^2 , σ_a^2 , and σ^2 .

(a) Show that if the means of the d_r , a_c , and ε_{rc} were in fact non-zero, they could not be distinguished from μ . Give a careful interpretation of μ .

(b) By arguing conditionally on the values of the d_r and a_c , show that the sums of squares for rows, $\sum_{r,c} (y_{rc} - \bar{y}_{..})^2$, columns, $\sum_{r,c} (\bar{y}_{.c} - \bar{y}_{..})^2$, and residuals, $\sum_{r,c} (y_{rc} - \bar{y}_{r.} - \bar{y}_{.c} + \bar{y}_{..})^2$, are independent. Obtain their distributions, and hence give formulae for unbiased estimates of the variances.

(c) The sums of squares for the analysis of variance table are 2969 for pipettes on 9 df, 2938 for doctors on 4 df, and 1176 for residual on 36 df. Obtain estimates of σ_d^2 , σ_a^2 , and σ^2 .

(d) Now suppose that general practitioners are to perform these measurements on a routine basis, with results referred to a central laboratory. Under the assumption that the data are normal, give the standard error for a measurement taken by a particular GP (i) if the apparatus is reusable, (ii) if a new set of apparatus must be used for each measurement. What standard error is appropriate if the measurements are rarely made, so that in effect

both GP and apparatus are new? What if the average of k measurements is recorded, and (i) apparatus is reusable, (ii) apparatus is not reusable?

- 4 Write down the linear mixed models corresponding to (9.8) and (9.11).
- 5 Use (3.21) and Exercise 8.5.2 to obtain (9.14) and (9.15).
- 6 On page 458, let $b^\dagger$ be any predictor of b based on y . Show that

$$\text{cov}(b, \tilde{b}) = \text{var}(\tilde{b}), \quad \text{cov}(\tilde{b}, y) = \text{cov}(b, y), \quad \text{cov}(b^\dagger, b) = \text{cov}(b^\dagger, \tilde{b}),$$

and deduce that

$$\text{corr}(b^\dagger, b)^2 = \text{corr}(b^\dagger, \tilde{b})^2 \text{corr}(\tilde{b}, b)^2.$$

Hence show that $\tilde{b}$ is the predictor of b that maximizes $\text{corr}(b^\dagger, b)$.

- 7 Consider applying the EM algorithm (Section 5.5.2) for estimation in a normal mixed model. Show that if the random effects b are treated as unobserved data, then the complete-data log likelihood is

$$-\frac{1}{2} \log |\Omega| - \frac{1}{2} (y - X\beta - Zb)^\top \Omega^{-1} (y - X\beta - Zb) - \frac{1}{2} \log |\Omega_b| - \frac{1}{2} b^\top \Omega_b^{-1} b,$$

and show that the only quantity needed for the M-step for estimation of components of Ω_b is $E(b^\top \Omega_b^{-1} b \mid y; \theta')$.

In the special case $\Omega_b = \sigma_b^2 I_q$, $\Omega = \sigma^2 I_n$, show that $E(b^\top b \mid y; \theta')$ equals

$$\text{tr} \left\{ \sigma_b^2 I_q - (\sigma_b^2 / \sigma')^2 Z^\top \Upsilon' Z \right\} + (\sigma_b^2 / \sigma')^4 (y - X\beta')^\top Z Z^\top \Upsilon' (y - X\beta').$$

Hence write down the form of the EM algorithm for this model.

(Searle *et al.*, 1992, Section 8.3)

- 8 Another approach to estimation in mixed models starts from noticing that

$$E\{(y - X\beta)(y - X\beta)^\top\} = \Omega + Z\Omega_b Z^\top$$

is linear in the variance parameters. Thus given an estimate $\hat{\beta}$, we could stack the unique elements of $(y - X\hat{\beta})(y - X\hat{\beta})^\top$ as a vector, v , say, and estimate the variance parameters by least squares regression of v on the appropriate design matrix. We then take $\hat{\beta} = (X^\top \Upsilon X)^{-1} X^\top \Upsilon y$, where Υ is formed using the variance estimates, and iterate the procedure.

Give the details of this for Example 9.16, using as initial value $\sigma_b^2 = 0$.

What difficulties do you see with this approach in general? Say how they might be overcome.

This is sometimes called iterative generalized least squares or IGLS estimation.

Frank Yates (1902–1994) was born in Manchester and educated there and in Cambridge. After working on a survey in Ghana he became Fisher's assistant at Rothamsted Experimental Station, where he rapidly became head of the statistics department. He made fundamental contributions to the design and analysis of experiments and to sample surveys. He quickly saw the importance of computing: in the 1950s he and his colleagues wrote machine code programs for analysis of variance and for survey analysis.

9.5 Bibliographic Notes

Designed experiments were used in the nineteenth century and earlier, but R. A. Fisher was the first to realise the importance of randomization, and his ideas had a strong impact from the 1920s onwards. His 1935 book on design of experiments, re-issued as part of Fisher (1990), is fundamental reading. Important further developments, particularly in agricultural experimentation, were due to F. Yates, with Yates (1937) highly influential. An excellent recent account is Cox and Reid (2000), which contains a full treatment of the topics of this chapter and other topics not mentioned here, with many further references. A more elementary discussion is Cobb (1998). Older standard texts are Cochran and Cox (1959) and the excellent non-mathematical treatment of Cox (1958).

The study of causality is central to scientific thought, but has been little discussed by statisticians until fairly recently. A valuable account and excellent starting-point for further reading is Chapter 8 of Edwards (2000), while Holland (1986) is a good review making links to the philosophical study of causation. Cox (1992) and Section 8.7 of Cox and Wermuth (1996) give a somewhat different perspective. Contrasting views on the usefulness of counterfactuals are held by Dawid (2000), Lauritzen (2001), and Pearl (2000).

Scheffé (1959) is a standard account of the analysis of variance.

Box *et al.* (1978) and Fleiss (1986) respectively discuss industrial experimentation and medical studies. Atkinson and Donev (1992) give a clear discussion of optimal experimental design; see also Silvey (1980) for a more theoretical account.

Components of variance models originated in astronomy in the 1860s and have been rediscovered and renamed many times since, being also known as hierarchical or multilevel models. Chapter 2 of Searle *et al.* (1992) gives a brief history oriented towards biometry and agriculture, while Goldstein (1995) describes their use in the social sciences, using slightly different estimation techniques and with a largely disjoint set of references! Although R. A. Fisher had discussed components of variance in the 1920s and 1930s, important work by Henderson (1953) and Hartley and Rao (1967) was key in a more general reformulation, while Patterson and Thompson (1971) built on earlier work to give a general discussion of REML estimation. Robinson (1991) is a passionate advocate of best linear unbiased prediction, with an interesting and wide-ranging discussion; see particularly the contribution by T. P. Speed. McCulloch and Searle (2001) give a recent account of variance components estimation in linear and generalized linear models.

9.6 Problems

- Example 9.6 is a *two-way layout with replication*, in which the j th replicate in row r and column c is

$$y_{rcj} = \mu + \alpha_r + \beta_c + \gamma_{rc} + \varepsilon_{rcj}, \quad r = 1, \dots, R, \quad c = 1, \dots, C, \quad j = 1, \dots, k.$$

The α_r and β_c represent the main effects of rows and columns; the γ_{rc} are row $\times$ column interactions; and $\varepsilon_{rcj} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$.

(a) Explain why an external estimate of σ^2 is needed if the γ_{rc} are known not to be constant, and $k = 1$.

(b) A first step in the analysis of such data is to calculate the cell mean and sums of squares $\bar{y}_{rc\cdot}$ and $\sum_j (y_{rcj} - \bar{y}_{rc\cdot})^2$. Show that the distribution of each cell sum of squares is $\sigma^2 \chi_{k-1}^2$, and explain what you might expect to learn from a plot of $\log \sum_j (y_{rcj} - \bar{y}_{rc\cdot})^2$ against $\log y_{rc\cdot}$. What does this plot show for the poisons data?

(c) The analysis of variance for this design is in Table 9.30. Show that

$$E \left\{ \sum_{r,c,j} (\bar{y}_{r\cdot\cdot} - \bar{y}_{\cdot\cdot\cdot})^2 \right\} = (R-1)\sigma^2 + kC \sum_r (\alpha_r - \bar{\alpha} + \bar{\gamma}_r - \bar{\gamma}_{\cdot\cdot})^2,$$

and write down $E\{\sum_{r,c,j} (\bar{y}_{rc\cdot} - \bar{y}_{\cdot\cdot\cdot})^2\}$. Explain why these depend on the α_r and β_c only through $\alpha_r - \bar{\alpha}$ and $\beta_c - \bar{\beta}$.

Table 9.30 Analysis of variance for two-way layout with replication.

Terms	df	Sum of squares
Rows	$R - 1$	$\sum_{r,c,j} (\bar{y}_{r..} - \bar{y}_{...})^2$
Columns	$C - 1$	$\sum_{r,c,j} (\bar{y}_{.c.} - \bar{y}_{...})^2$
Rows $\times$ Columns	$(R - 1)(C - 1)$	$\sum_{r,c,j} (\bar{y}_{rc.} - \bar{y}_{r..} - \bar{y}_{.c.} + \bar{y}_{...})^2$
Residual	$RC(k - 1)$	$\sum_{r,c,j} (y_{rcj} - \bar{y}_{rc.})^2$

(d) Show that

$$E \left\{ \sum_{r,c,j} (\bar{y}_{rc.} - \bar{y}_{r..} - \bar{y}_{.c.} + \bar{y}_{...})^2 \right\} = (R - 1)(C - 1)\sigma^2 + k \sum_{rc} (\gamma_{rc} - \bar{\gamma}_{r.} - \bar{\gamma}_{.c} + \bar{\gamma}_{..})^2.$$

Under what circumstances does this equal $(R - 1)(C - 1)\sigma^2$?

- 2 Let y_{gr} , $g = 1, \dots, G$, $r = 1, \dots, R$, be independent normal random variables with means μ_{gr} and common variance σ^2 .

(a) Assume the one-way analysis of variance model, namely that $\mu_{gr} = \mu_g$, so that the y_{gr} are replicate measurements with the same mean, and find the sufficient statistics for the μ_s and σ^2 . Show that these are equivalent to

$$\bar{y}_{1.}, \dots, \bar{y}_{G.}, \quad SS = \sum_{g=1}^G \sum_{r=1}^R (y_{gr} - \bar{y}_{g.})^2,$$

where $\bar{y}_{g.} = R^{-1} \sum_{r=1}^R y_{gr}$; note that

$$\sum_r (y_{gr} - \mu_g)^2 = \sum_r (y_{gr} - \bar{y}_{g.})^2 + R(\bar{y}_{g.} - \mu_g)^2.$$

Find the distribution of $\sum_{r=1}^R (y_{gr} - \bar{y}_{g.})^2$.

(b) Prove that SS is independent of the group means, and that it is proportional to a chi-squared random variable on $G(R - 1)$ degrees of freedom.

(c) Let $\bar{y}_{..} = G^{-1} \sum_g \bar{y}_{g.}$ denote the overall mean. If $\mu_1 = \dots = \mu_G$, show that the distribution of $SS_G = R \sum_{g=1}^G (y_{g.} - \bar{y}_{..})^2$ is proportional to a chi-squared distribution on $G - 1$ degrees of freedom. Hence find the distribution of $G(R - 1)SS_G / (G - 1)S^2$, when the means are equal.

(d) Samples of the same material are sent to four laboratories for chemical analysis as part of a study to determine whether laboratories give the same results. The results for laboratories A–D are:

A	58.7	61.4	60.9	59.1	58.2
B	62.7	64.5	63.1	59.2	60.3
C	55.9	56.1	57.3	55.2	58.1
D	60.7	60.3	60.9	61.4	62.3

$$F_{3,16}(0.95) = 3.24.$$

Test the hypothesis that the means are different and comment.

- 3 (a) For $n = 2m + 1$ and positive integer m , suppose that $y_1, \dots, y_n$ follow the normal linear model

$$y_j = \beta_0 + \sum_{k=1}^m \{\beta_k \cos(2\pi k j / n) + \gamma_k \sin(2\pi k j / n)\} + \varepsilon_j.$$

Show that the last $2m$ columns of the design matrix for this model are orthogonal contrasts, and find the least squares estimators of the parameters.

(b) Show that the overall sum of squares $\sum y_j^2$ may be split into a component $n\bar{y}^2$ corresponding to the grand mean, and m components $I_j = n(\beta_j^2 + \gamma_j^2)/2$ corresponding to variation with frequency $2\pi j/n$, $j = 1, \dots, m$. Show that the I_j are independent, and

that if there is no cyclical variation, $\frac{1}{2}I_j$ has an exponential distribution with mean σ^2 , whatever the value of n .

(c) Dataframe `venice` contains the annual maximum tides at Venice for the 51 years 1931–1981. It has been suggested that they may vary according to the astronomical tidal cycle, which has period 18.62 years, and that they may also be affected by the sunspot cycle, whose period is 11 years. To assess this:

```
attach(venice)
split.screen(c(1,2))
screen(1); plot(year,sea,ylab="Sea level (cm)")
n <- 51;
k1 <- 19; omega1 <- 2*pi*k1/n # roughly 18.62 years
k2 <- 11; omega2 <- 2*pi*k2/n
X19 <- cbind(sin(year*omega1),cos(year*omega1))
X11 <- cbind(sin(year*omega2),cos(year*omega2))
crossprod(cbind(rep(1,n), X11,X19)) # matrix X^T X
venice.lm <- lm(sea~X19+X11)
anova(venice.lm,test="F")
```

Do these cycles seem to be present? How would the knowledge that the errors are non-normal affect your conclusion?

Hint: If $\omega = 2\pi p/n$, where p is an integer in the range $1, \dots, [n/2]$, and k_1, k_2 are integers in the range $1, \dots, n$, then

$$\sum_{j=1}^n \cos(k_1 \omega j) = \sum_{j=1}^n \sin(k_1 \omega j) = \sum_{j=1}^n \cos(k_1 \omega j) \sin(k_2 \omega j) = 0,$$

$$\sum_{j=1}^n \cos(k_1 \omega j) \cos(k_2 \omega j) = \sum_{j=1}^n \sin(k_1 \omega j) \sin(k_2 \omega j) = \begin{cases} n/2, & k_1 = k_2, \\ 0, & \text{otherwise.} \end{cases}$$

To verify this, consider the real and imaginary parts of the sums $e^{ik_1\omega} + \dots + e^{ik_1\omega n}$ and $e^{i(k_1+k_2)\omega} + \dots + e^{i(k_1+k_2)\omega n}$.

- 4 (a) Suppose that times had been obtained only for the odd-numbered setups Example 8.4. Further suppose that two people had been involved, and that one had ridden the bike for those setups with the seat in the low position, while the other had done so for those with the seat in the high position. Show that in this case the estimated seat and person effects are the same, and discuss what this implies about what the seat height should be.
- (b) Now suppose that one person had ridden the bike for setups 1, 7, 9, and 15, while the other person had done so for setups 3, 5, 11, and 13. Show that provided there are no second- and higher-order interactions, the seat, tyre, dynamo and person effects could all be estimated from this experiment. What problem would arise if there was known to be an interaction of seat and dynamo? Explain how to modify the design to overcome this if it is known that there are no other second-order interactions.
- 5 The 3×3 Latin square laid out as

A	B	C
B	C	A
C	A	B

has nine observations classified by rows, columns, and the treatments A, B, and C. The corner-point parametrization for the means is

$$\begin{array}{ccc} \mu & \mu + \alpha_1 + \gamma_1 & \mu + \alpha_2 + \gamma_2 \\ \mu + \delta_1 + \gamma_1 & \mu + \alpha_1 + \delta_1 + \gamma_2 & \mu + \alpha_2 + \delta_1 + \gamma_2 \\ \mu + \delta_2 + \gamma_2 & \mu + \alpha_1 + \delta_2 + \gamma_1 & \mu + \alpha_2 + \delta_2 + \gamma_1 \end{array} ;$$

α_1, α_2 are column effects, δ_1, δ_2 are row effects and γ_1, γ_2 are treatment effects.

Write out the corresponding design matrix and verify that the effects of rows, columns, and treatments are not orthogonal. Check that the matrix is orthogonal when the terms for rows, columns and treatments are centred. Without doing any calculations, say whether

the same is true for the *Graeco-Latin square*

$$\begin{array}{ccc} A\alpha & B\beta & C\gamma \\ B\gamma & C\alpha & A\beta \\ C\beta & A\gamma & B\alpha \end{array}$$

in which there is the further set of treatments α, β, γ .

- 6 (a) Let A be a $q \times q$ positive definite symmetric matrix, and let b and y be two random variables with a joint distribution. Let $b^\dagger = b^\dagger(y)$ denote any predictor of b based on y , and let $\tilde{b} = E(b | y)$, assuming this is finite for all y . By writing

$$E\{(b^\dagger - b)^T A (b^\dagger - b)\} = E_y E_{b|y}\{(b^\dagger - \tilde{b} + \tilde{b} - b)^T A (b^\dagger - \tilde{b} + \tilde{b} - b) | y\},$$

deduce that this is minimized when $b^\dagger = \tilde{b}$, for any A and any joint distribution. Note that $E(\tilde{b}) = E(b)$, so in this sense $\tilde{b}$ is unbiased.

(b) Now consider the class of linear predictors $b^\dagger(y) = a + By$, where a and B are constants of dimensions $q \times 1$ and $q \times n$. Let $W = b - By$, and show that

$$E\{(b^\dagger - b)^T A (b^\dagger - b)\} = \{a - E(W)\}^T A \{a - E(W)\} + \text{tr}\{A \text{Var}(W)\}.$$

Deduce that this is minimized by taking $a = -BX\beta$ and $B = \sigma^{-2}\Omega_b Z^T \Upsilon$. Hence show that $\sigma^{-2}\Omega_b Z^T \Upsilon(y - X\beta)$ is the best linear predictor of b whatever the distributions of b and y .

You may like to know that
 $|cI_k + d1_k 1_k^T| =$
 $c^{k-1}(c + kd).$

- 7 (a) Show that the log likelihood in Example 9.17 is

$$\ell(\mu, \sigma^2, \psi) \equiv -\frac{1}{2\sigma^2} \left\{ SS_w + \sum_{i=1}^q \frac{n_i(\bar{y}_{i.} - \mu)^2}{1 + n_i\psi} \right\} - \frac{n}{2} \log \sigma^2 - \frac{1}{2} \sum_{i=1}^q \log(1 + n_i\psi),$$

where $\psi = \sigma_b^2/\sigma^2$ and $SS_w = \sum_{i,j} (y_{ij} - \bar{y}_{i.})^2$.

- (b) Show that using REML increases the log likelihood by

$$\frac{1}{2} \log \sigma^2 - \frac{1}{2} \log \left\{ \sum_{i=1}^q \frac{n_i}{1 + n_i\psi} \right\}.$$

- (c) Show that if ψ is known, then

$$\hat{\mu}(\psi) = \frac{\sum_{i=1}^q n_i \bar{y}_{i.} / (1 + \psi n_i)}{\sum_{i=1}^q n_i / (1 + \psi n_i)},$$

and deduce that $\hat{\mu} = \bar{y}_{..}$ if the design is balanced, that is, $n_1 = \dots = n_q$. In this case obtain $\hat{\sigma}^2(\psi)$ and $\hat{\psi}$ and compare with the results for (9.8). What is the effect of using REML?